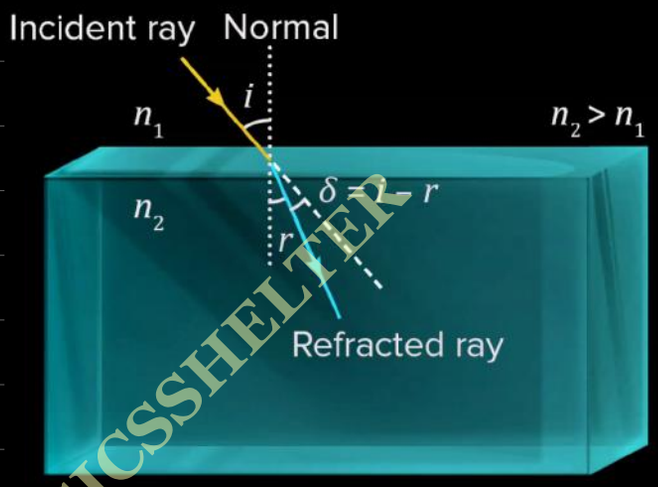
Refractive Index: Ratio of speed of light in free space to that in a given medium is called refractive index. Mathematically,

$$\mu = c/v$$

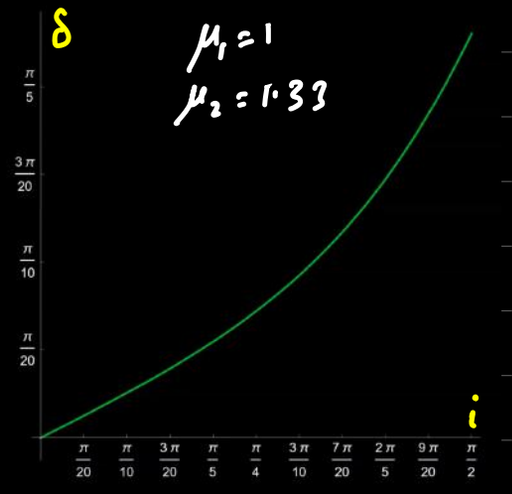
Here $\mu \rightarrow$ Refractive index, $v \rightarrow$ speed in the given medium.

Snell's Law:

Whenever a light rays enters from medium 1 to medium 2, it gets refracted such that $\mu_1 \sin i_1 = \mu_2 \sin i_2$ (or $n_1 \sin i_1 = n_2 \sin i_2$). See adjacent figure. Snell's law can derived using Fermat's principle i.e. light takes a stationary path while moving from one point to another (i.e minimum time, or degenerate equal time for most cases).

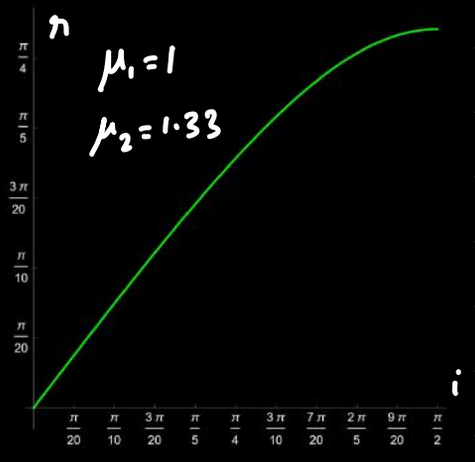


The angle between incident ray and refracted ray is called angle of deviation. A typical graph of angle of deviation from air in to water is shown in adjacent figure.



A similar graph is also obtained for entry from glass to air however after a certain angle of incidence, the light undergoes Total internal reflection. ($i_c = \sin^{-1}(1/\mu)$).

A graph of i vs r is shown in the next figure for entry from rarer to a denser medium. In case of entry from denser to rarer medium this graph will be flipped about the line $y = x$. (Recall principle of reversibility).



→ When a ray moves from rarer medium to a denser medium, it deflects towards the normal

→ When a ray deflects from denser medium to

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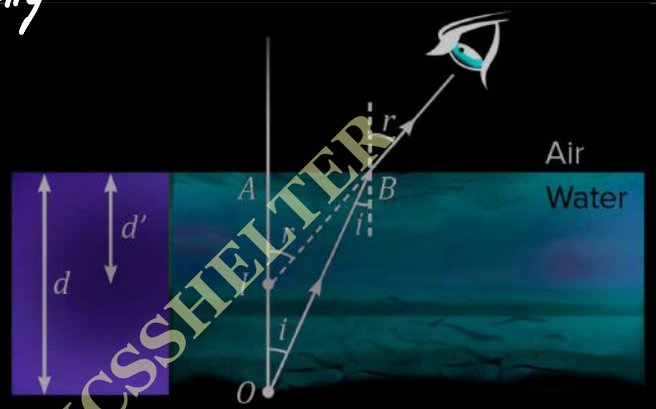
rarer medium, it deflects away from normal.

→ Normal, incident ray and reflected ray are in same plane.

→ Normal in Geometrical optics means normal to the surface of constant refractive index, or the direction of steepest change in refractive index.

APPARENT DEPTH AT NEAR NORMAL INCIDENCE:

Suppose we are viewing an object normally such that rays entering our eyes make small angles with the normal to a flat refracting surface, then we get ray diagram as shown in adjacent figure with i and r being small angles. Now $d' = AB\cot r$ and $d = AB\cot i$ thus:



$$\frac{d'}{d} = \frac{\cot r}{\cot i} = \frac{\tan i}{\tan r} \approx \frac{\sin i}{\sin r}$$

Also $\mu \sin i = 1 \sin r \Rightarrow \sin i / \sin r = 1/\mu$

Thus $\boxed{d' = d/\mu}$

→ In case instead of air we had some other liquid with refractive index μ_l then the apparent depth of image from surface would have been:

$$d' = \frac{d\mu_l}{\mu_w}$$ where μ_w is refractive index of water

and μ_l is refractive index of liquid, we could also write this expression as:

$$\boxed{d' = \frac{d}{\mu_{wl}}}$$ where μ_{wl} is relative refractive index of water wrt liquid i.e μ_w/μ_l. If $\mu_l < \mu_w$ then $d' < d$ and if $\mu_l > \mu_w$ then $d' > d$ image of the object will be formed at a deeper level.

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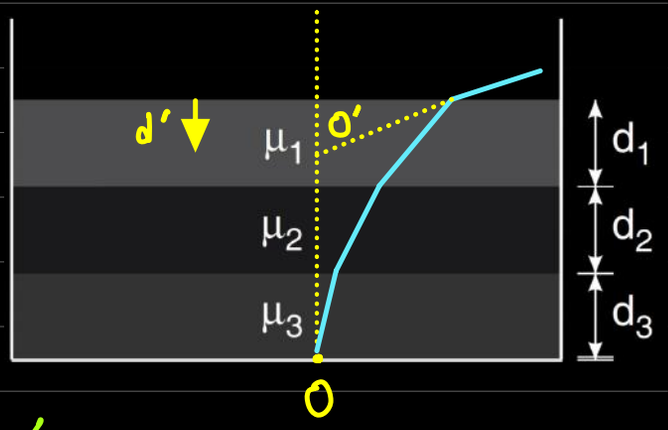
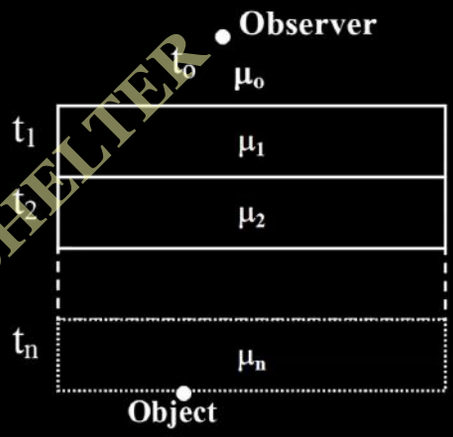
→ In case of several different media,

$$d' = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \frac{d_3}{\mu_3}$$



→ In case the observation medium has refractive index μ_0 and thickness t_0. Then apparent distance of image from observer is simply:

$$d_{app} = \mu_0 \left[\frac{t_0}{\mu_0} + \frac{t_1}{\mu_1} + \frac{t_2}{\mu_2} + \dots \right]$$



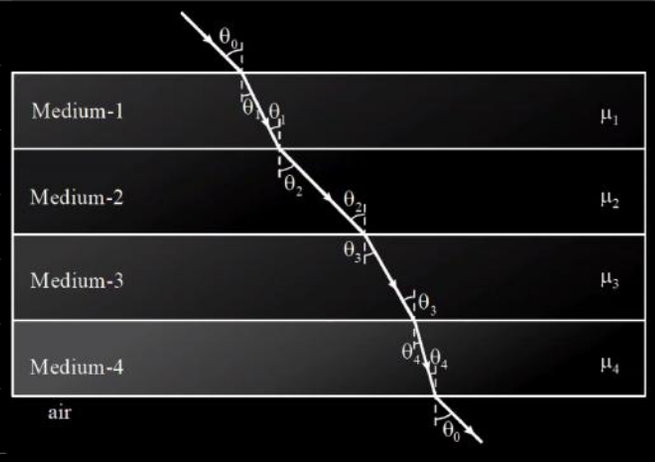
In case of variable refractive index:

$$d_{app} = \mu_0 \int \left[\frac{dt}{\mu} \right]$$

System Of Parallel Slabs With Same Initial And Final Mediums:

Whenever we have a system of surfaces where all surface normals are parallel to each other then we can write $\mu_0 \sin\theta_0 = \mu_1 \sin\theta_1 = \mu_2 \sin\theta_2 = \dots = \mu_0 \sin\theta_0$

Thus exiting ray will be parallel to the entry ray in case final and initial media are same. (This would not be true if surface normals were not // to each other. Because then angle of refraction at entry will not be equal to angle of incident while exiting the medium.



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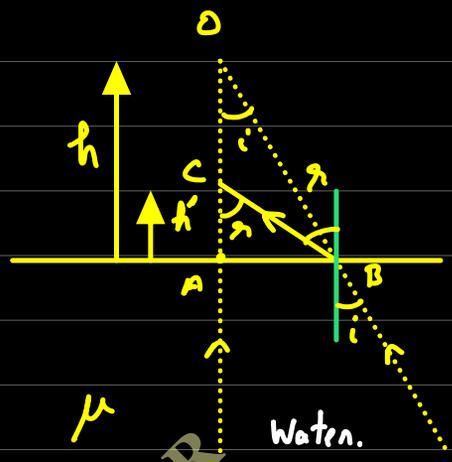
Convergence Point Distance Due To Refracting Surface:



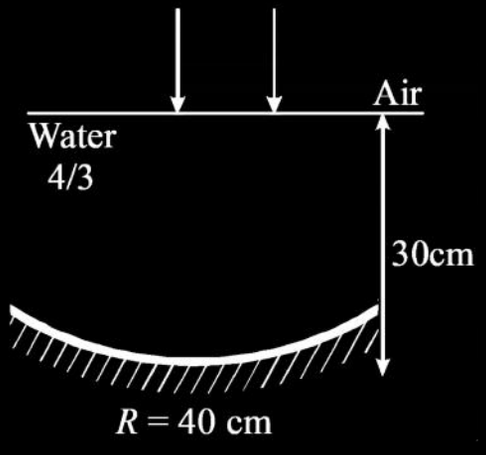
Let h be the convergence distance of a beam had it not refracted at a refracting surface and let h' be the convergence distance due to refraction then:

$$\frac{h'}{h} = \frac{\cot r}{\cot i} = \frac{\tan i}{\tan r} \approx \frac{\sin i}{\sin r}$$

Also $\mu \sin i = 1 \sin r$, thus $\sin i / \sin r = \mu$ thus,

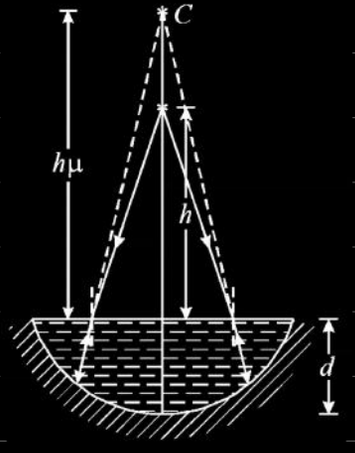
$$\frac{h'}{h} = \frac{1}{\mu} \Rightarrow \boxed{h' = h/\mu}$$, Had the ray been going in to some liquid instead of water then $h' = h/\mu_{wl}$ where $\mu_{wl} = \mu_w/\mu_l$ μ_w: Refractive index of water, μ_l: Refractive index of liquid.

Example: A concave mirror is placed in water and a parallel beam of light falls on it as shown in the figure. What is the final image formed?



Solution: Focal length of mirror : -20 cm
thus, after processing from mirror the image is formed 10 cm below the surface.
Thus its depth below surface: $h = 10$ cm
Now apparent depth: $h' = h/\mu = 10/(4/3)$
$\Rightarrow h' = 7.5$ cm below the surface. Ans.

Example: A concave mirror of focal length F is filled with a liquid of refractive index μ up to a depth d. Where should a pin be placed on its principal axis for auto collimation.



Solution: For auto collimation, the rays should fall normally on the mirror ∴ intermediate image

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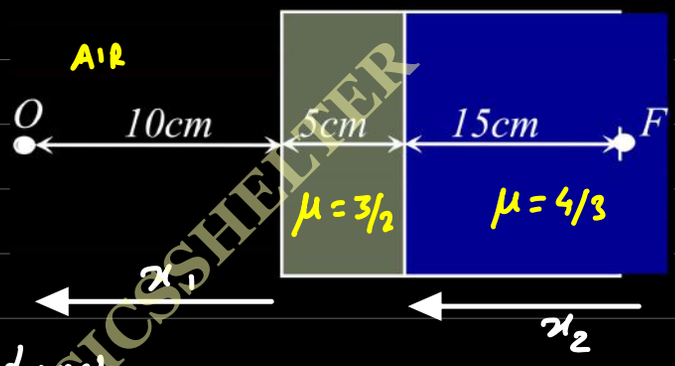
after refraction from upper surface must be formed at centre of curvature of the mirror. Now apparent distance of pin from surface as seen from the pole of the mirror:

$$x_{app} = \mu\left[\frac{d}{\mu} + \frac{h}{1}\right] = \mu h + d = 2|F|$$

Thus, $h = [2|F| - d]/\mu$ ans.

EXAMPLE: In the shown system, O is a human observes and F is a fish. (a) What is apparent distance of fish as seen by the human (b) What is the apparent distance of human as seen by the fish.



(c) Suppose fish is swimming with 2 cms^{-1} towards human and human is walking at 1 cms^{-1} towards fish, what is the apparent velocity of the human from the fish frame?

Solution: For human, apparent distance, $x_h = 1\left[\frac{10}{1} + \frac{5}{3/2} + \frac{15}{(4/3)}\right]$

$\Rightarrow x_{app} = 295/12$ cm ans., Similarly for fish, apparent distance of human:

$$x_f = \frac{4}{3}\left[\frac{15}{4/3} + \frac{5}{3/2} + \frac{10}{1}\right] = \frac{295}{9} \text{ cm ans.}$$

(c) Apparent left coordinate of human as seen from fish frame:

$$x_a = \frac{4}{3}\left[\frac{x_2}{4/3} + \frac{5}{3/2} + \frac{x_1}{1}\right]$$

$\Rightarrow$ Apparent left velocity $= \dot{x}_a = \frac{4}{3}\left[\frac{3}{4}\dot{x}_2 + \dot{x}_1\right] = \frac{4}{3}\left[\frac{3}{4}\times -2 + -1\right]$

$\Rightarrow \dot{x}_a = -10/3$ cms^{-1} $\Rightarrow$ man appears to move with $10/3$ cms^{-1} ans.

→ Think: suppose water is draining out from a an aquarium and fish is moving upwards. How will you find apparent velocity of fish as seen from above.

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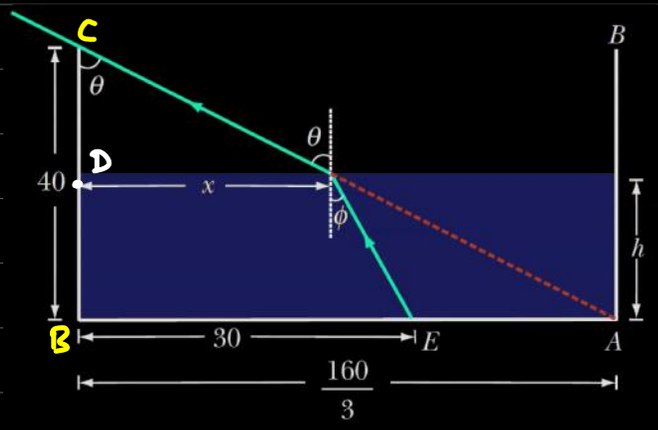
EXAMPLE: An observer is viewing along the green line as shown in the figure. Without any water filled, he is able to see point A along this line of sight. When water is filled to a depth h, he is able to see point E along that direction. Find θ, ϕ and depth to which water is filled. $\mu_w = 4/3$.



Sol: $\tan\theta = AB/CB = (160/3)/40 = 4/3 \Rightarrow \theta = 53^\circ$ Ans.

Now $1\sin\theta = \frac{4}{3}\sin\phi \Rightarrow \sin\phi = \frac{3}{4}\times\frac{4}{5} = \frac{3}{5} \Rightarrow \phi = 37^\circ$ Ans.

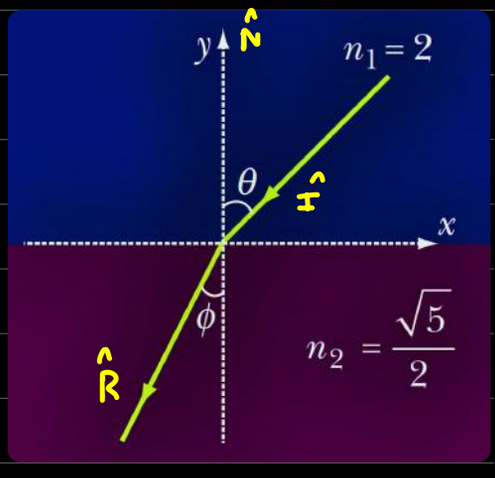
Now $x + h\tan 37^\circ = 30 \Rightarrow x + \frac{3}{4}h = 30$ ①

also $\frac{x}{40-h} = \tan 53^\circ = 4/3 \Rightarrow 3x = 160 - 4h$ ②

Solving ① & ②: $x = 0$ $h = 40\,cm$ Ans.
(i.e tank is completely filled)

FINDING REFRACTED RAY IN TERMS OF INCIDENT AND NORMAL UNIT VECTORS:

Consider the refraction shown in adjacent figure.



Here $\hat{R} = \cos\phi(-\hat{N}) + \sin\phi(-\hat{i})$

$= \cos\phi(-\hat{N}) + \mu_{12}\sin\theta(-\hat{i})$

$= \cos\phi(-\hat{N}) + \mu_{12}(\hat{N}\times\hat{I})\times\hat{N}$

Now $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1-\mu_{12}^2\sin^2\theta}$

Also $(\hat{N}\times\hat{I})\times\hat{N} = -[\hat{N}\times(\hat{N}\times\hat{I})] = -[\hat{N}(\hat{N}\cdot\hat{I}) - \hat{I}]$

Thus $\hat{R} = -\sqrt{1-\mu_{12}\sin^2\theta}\,\hat{N} - \mu_{12}[\hat{N}(\hat{N}\cdot\hat{I}) - \hat{I}]$

$\hat{R} = \mu_{12}\hat{I} - \hat{N}[\mu_{12}\hat{N}\cdot\hat{I} - \sqrt{1-\mu_{12}\sin^2\theta}]$

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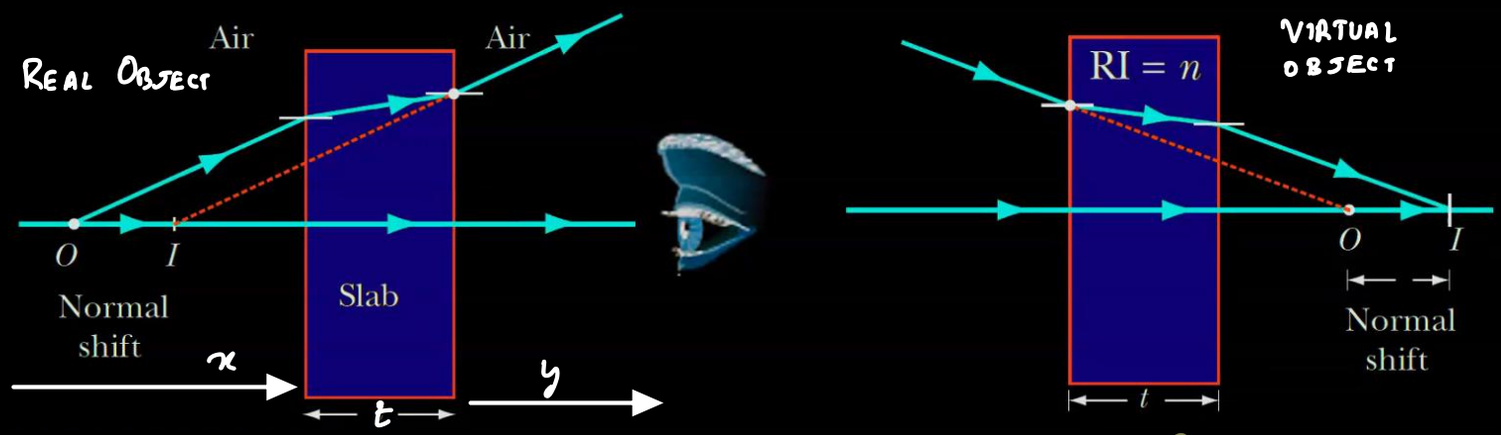
Normal Shift From A Glass Slab:



Whenever an object is viewed through a glass slab, it appears nearer by a distance $t(1-1/\mu)$. This can be easily seen using apparent distance formula. Consider the first figure above. Here apparent distance is:

$$1\left[\frac{y}{1}+\frac{t}{\mu}+\frac{x}{1}\right] = x+y+\frac{t}{\mu}$$

while the actual distance is $x+y+t$. Thus the apparent distance is shorter by an amount $t(1-1/\mu)$. Similar shift will also be observed with a converging beam as shown in figure 2.

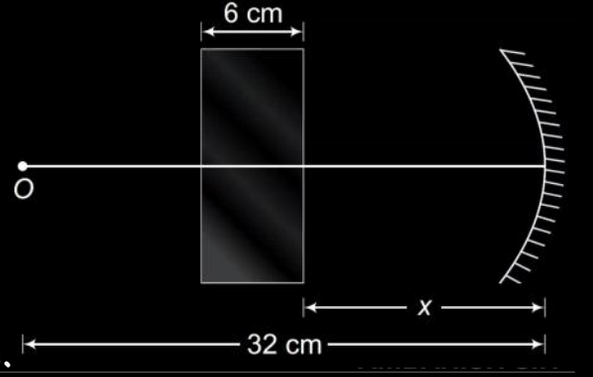
→ The shift is along the direction of incident light and given by

$$\boxed{\Delta x = t\left[1-\frac{1}{\mu}\right]}$$

→ If the observation medium were not air then we will have to replace the refractive index of slab by relative refractive index.

→ If the image is formed after crossing the last refracting surface, it is real while if it is formed before that, it is virtual (Formed by divergent rays).

Example: A point object O is placed in front of a concave mirror of focal length 10cm. A glass slab of refractive index $\mu = 3/2$ and thickness 6cm is inserted between the object and the mirror. Find the final position and nature of image if the distance x is (a) 5cm (b) 20cm.



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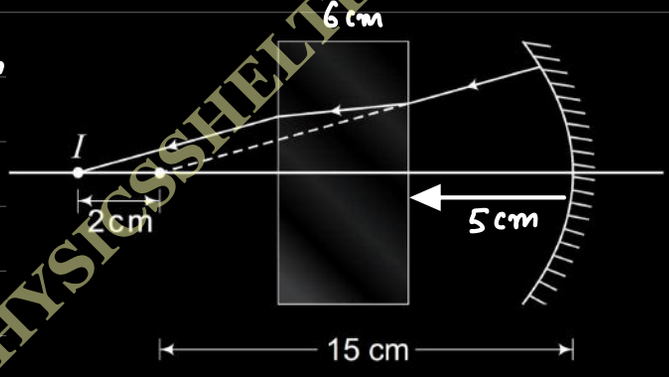
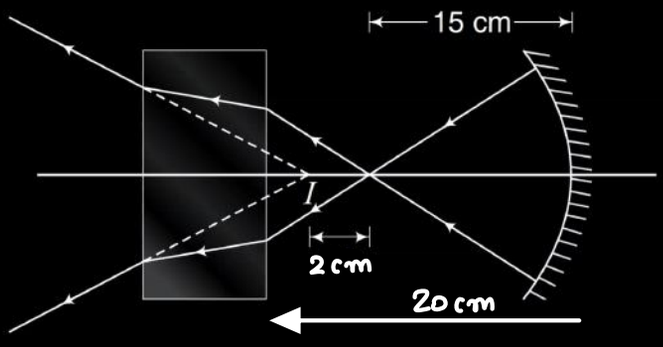
Solution: The rays are first processed by the slab, then by the mirror and then once again by the slab. Now, slab creates a normal shift of $6(1-1/(3/2)) = 2$ cm. Thus for mirror, $u = -30$ cm. Now applying mirror formula:

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{-30}+\frac{1}{v}=\frac{1}{-10} \Rightarrow v = -15 \text{ cm}.$$ Now these rays are once again processed by the slab to produce a normal shift of 2 cm in the reversed direction. Thus final image is formed at 15+2 i.e 17 cm from the pole of the mirror.

Now when slab is at a distance of 5 cm from the mirror, this image is formed after crossing the last surface of the slab, hence it is a real image (see adjacent figure).



With slab at a distance of 20 cm, the final image is virtual see adjacent figure. Here the final image is formed before crossing the last refracting surface.



Think: What will be the nature of final image if slab were at a distance of 14 cm?

Sol: Now the final image will be formed inside the slab, however still it is formed before crossing the last refracting surface, thus image will be virtual.

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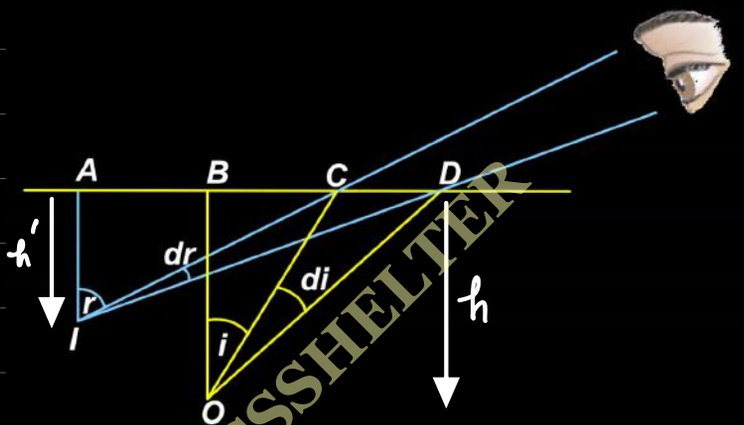
EXAMPLE: IRODOV: 5.18:

A man standing on the edge of a swimming pool looks at a stone lying on the bottom. The depth of the swimming pool is equal to h. At what distance from the surface of water is the image of the stone formed if the line of vision makes an angle θ with the normal to the surface?



Solution: Consider the adjacent ray diagram. Here we have replaced θ with symbol r because it is same as angle of refraction. Here we are considering image formation from two very close rays with incidence angles i and $i+di$ and corresponding refraction angles r and $r+dr$.

Now consider AIC and AID, we can write the length CD using these triangles as $CD = h' \tan(r+dr) - h' \tan r = h' \sec^2 r\, dr$ ①

Now consider triangles OBC and OBD, using these we can write the length CD as: $CD = h \tan(i+di) - h \tan i = h \sec^2 i\, di$ ②

Thus using ① and ② we can write:

$$h' \sec^2 r\, dr = h \sec^2 i\, di \quad ③$$

Also $\mu \sin i = \sin r \Rightarrow \mu \cos i \frac{di}{dr} = \cos r$

Thus $\cos r\, dr = \mu \cos i\, di$ ④ Now ③/④ gives:

$$h' \sec^3 r = \frac{h}{\mu} \sec^3(i) \Rightarrow \boxed{h' = \frac{h \cos^3 r}{\mu \cos^3 i}}$$

Now $r \to \theta$, $\cos i = \sqrt{1-\sin^2 i} = \sqrt{1-\frac{\sin^2\theta}{\mu^2}} = \sqrt{\mu^2 - \sin^2\theta}/\mu$

Thus $$\boxed{h' = \frac{\mu^2 h \cos^3\theta}{(\mu^2 - \sin^2\theta)^{3/2}}}$$

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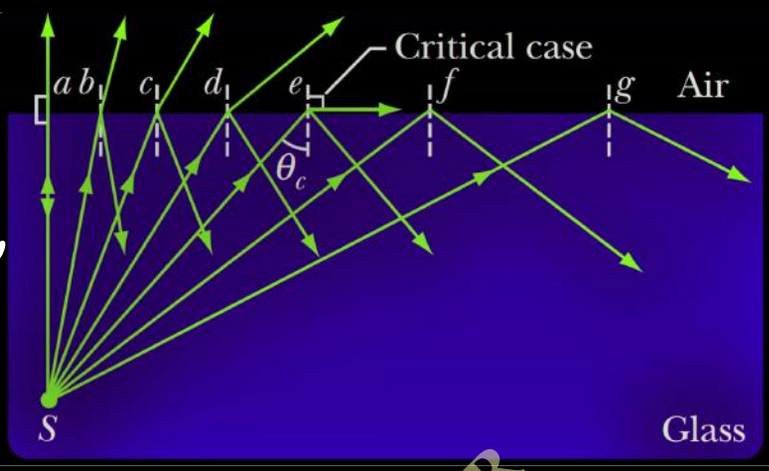
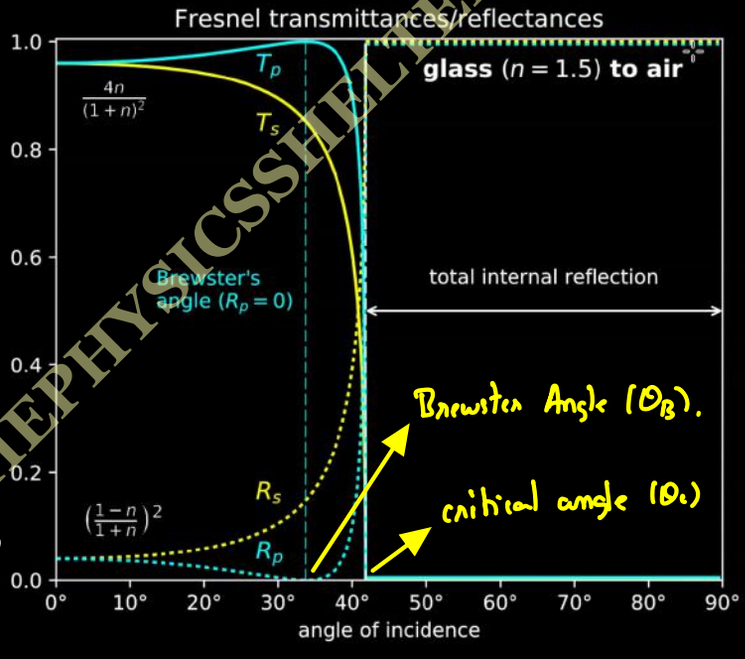
Total And Partial Reflection For A Ray Traveling From Denser To Rarer Medium:



Whenever a ray is incident from one medium to another a portion of it is reflected back and another portion is transmitted at angles of incidence less than a certain angle (known as critical angle). If the angle of incidence is greater than or equal to critical angle there is 100% reflection (Critical angle exists only for rays traveling from denser to rarer medium. The critical angle is the angle for which $r = 90°$, thus $\mu \sin\theta_c = 1 \sin 90° \Rightarrow \theta_c = \sin^{-1}(1/\mu)$ [For medium to air].



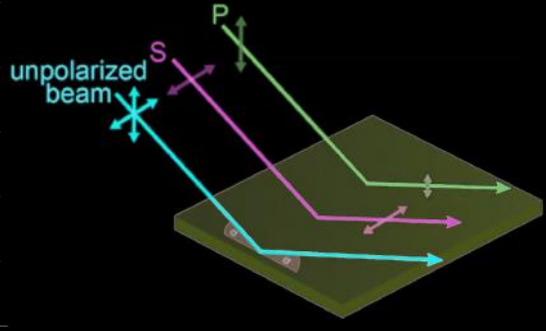
The adjacent figure gives the transmission and reflection coefficients variation for various angles of incidence. The two graphs are shown for two possible polarization directions:

p wave: The wave for which electric field oscillates in plane of incidence.

s wave: The wave for which electric field oscillate ⊥ to plane of incidence.

Apart from critical angle there is another important angle known as Brewster's angle at which 100% of s wave is transmitted. (This happens when reflected and refracted beams are mutually ⊥). You will know more about it in the chapter on polarization. ($\theta_B = \tan^{-1} 1/\mu$).

The s wave and p wave can be understood from the adjacent figure clearly. In general the unpolarized light can be thought to be containing both polarizations. Due to different transmissivities / reflectivities of the two components in general the transmitted and



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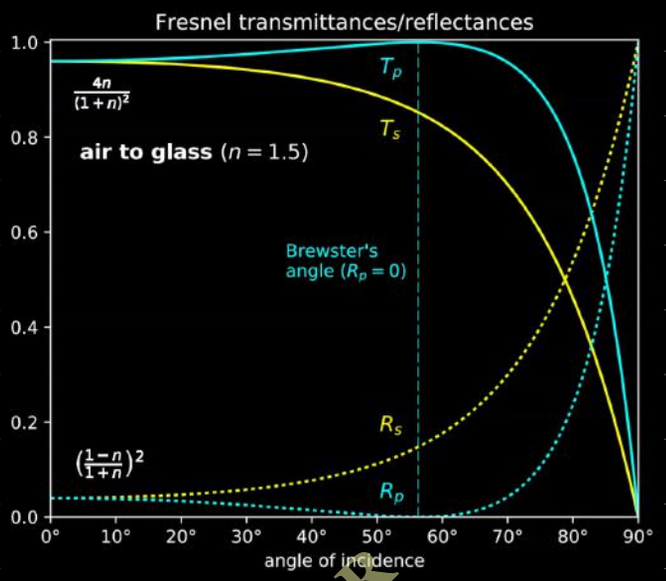
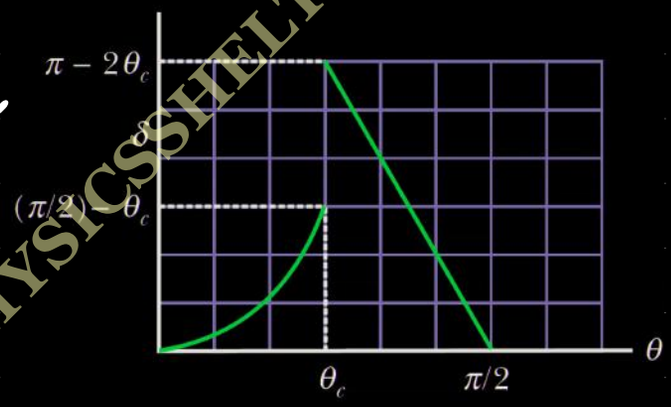
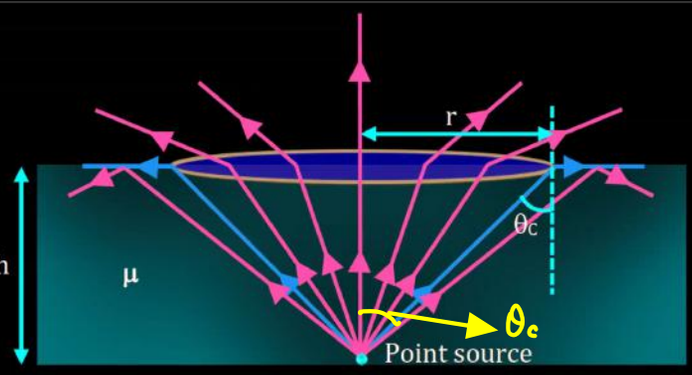
reflected rays are partially polarized. At Brewster's angle the reflected light is completely S-polarized. The reflectivity and transmissivity coefficients for a ray coming from a rarer medium to a denser medium are shown in adjacent figure. Here there is no critical angle but Brewster's angle is still there.



If we ignore the partial reflection before critical angle (This is generally done in most JEE problems), the graph of angle of deviation vs angle of incidence is as shown in adjacent figure. Here the abrupt change is because of TIR happening at critical angle.



EXAMPLE: A point isotropic source lies beneath a lake surface at a depth h. A disc is to be placed on the surface so that no light escapes the surface by refraction. Find the minimum required radius of the disc (see figure).

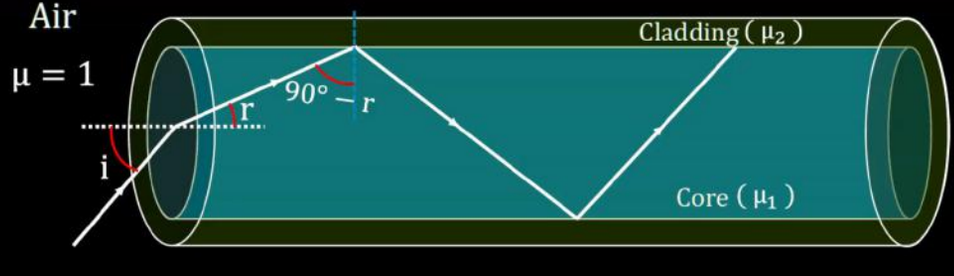


Solution: From the figure, $r_{min} = h \tan\theta_c$, Also $\sin\theta_c = 1/\mu$

$\Rightarrow \tan\theta_c = 1/\sqrt{\mu^2 - 1}$ thus required radius, $r = h/\sqrt{\mu^2 - 1}$ Ans.

EXAMPLE: FIBER OPTICS CABLE:

The adjacent figure shows a portion of fiber optic cable, find the maximum value of i if we want the signal ray to repeatedly



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undergo TIR in the cable.

Solution: Consider the ray diagram. From the diagram, it is clear that $90 - r > \theta_c \Rightarrow \cos r > \sin\theta_c = \mu_2/\mu_1$ i.e:

$$\cos r > \mu_2/\mu_1 \quad (1)$$

Also $1 \sin i = \mu_1 \sin r \Rightarrow \sin r = \frac{\sin i}{\mu_1} \Rightarrow \cos r = \sqrt{1-\left(\frac{\sin i}{\mu_1}\right)^2}$ (2)

Using (2) in (1):

$$\sqrt{1-\frac{\sin^2 i}{\mu_1^2}} > \frac{\mu_2}{\mu_1}$$

$$\Rightarrow 1-\frac{\sin^2 i}{\mu_1^2} > \frac{\mu_2^2}{\mu_1^2} \Rightarrow \mu_1^2 - \sin^2 i > \mu_2^2$$

$\Rightarrow \sin i < \sqrt{\mu_1^2-\mu_2^2}$, Here i is called minimum acceptance angle.

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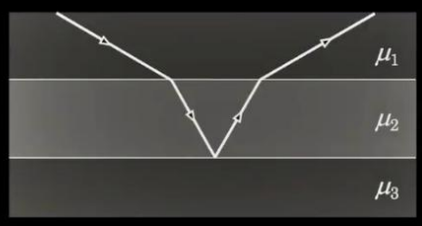
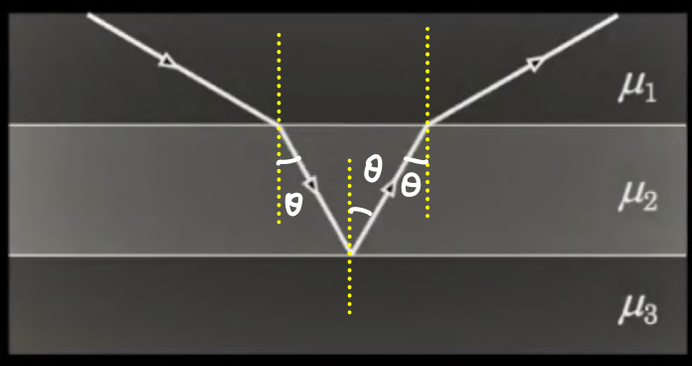
EXAMPLE: PATHFINDER MCQS:

Light travelling through three transparent substances follows a path shown in the figure. Arrange their indices of refraction in order from smallest to largest.



(a) $\mu_3 < \mu_1 < \mu_2$
(b) $\mu_1 < \mu_3 < \mu_2$
(c) Neither of the above is possible.
(d) More information is required.

Solution: From the ray diagram, $\mu_2 > \mu_1$ (∵ bending towards normal) and $\mu_2 > \mu_3$ (∵ TIR). Now we need to compare μ_1 and μ_3. We see that θ is more than critical angle for $\mu_2 - \mu_3$ interface while it is less than critical angle for $\mu_2 - \mu_1$ interface, i.e for $\mu_2 - \mu_1$ interface critical angle is larger. Now $\sin\theta_c = \frac{\mu_1}{\mu_2}$ for this interface while for lower interface, it is $\frac{\mu_3}{\mu_2}$, therefore $\mu_1 > \mu_3$. Thus (a) is the correct option.



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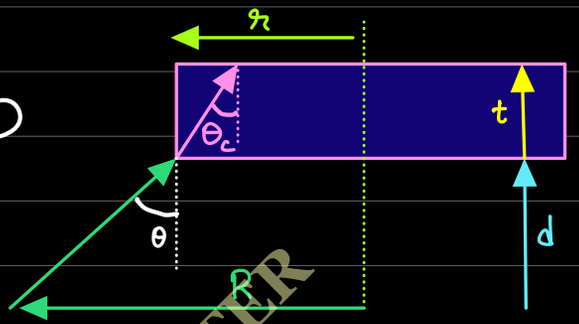
EXAMPLE: PATHFINDER: BUILD: 11

A research submarine has a circular glass window in its bottom to observe the seabed. Radius of the window is $r = 30$ cm, thickness of the glass is $t = 20$ mm, refractive index of water is $\mu_w = 1.25$ and that of the glass is $\mu_g = 1.5$. If the seabed is $d = 6.0$ m beneath the window, estimate radius of portion of the seabed visible through the window.



Solution: Consider the ray diagram in adjacent figure. The required radius $R = r + d\tan\theta$ ①

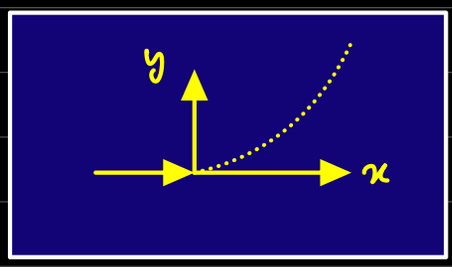
Now $\mu_g \sin\theta_c = 1$ ②

$$\mu_w \sin\theta = \mu_g \sin\theta_c \quad ②$$

$$\Rightarrow \sin\theta = \frac{1}{\mu_w} \Rightarrow \tan\theta = 1/\sqrt{\mu_w^2 - 1}$$

Thus $R = r + d/\sqrt{\mu_w^2 - 1} \approx 8.3$ m Ans.

EXAMPLE: A ray of light is travelling almost horizontally in a medium whose refractive index is increasing with y according to function $\mu(y)$. If μ_0 is the refractive index at the origin find y" of its path at a general point in terms of $\mu(y)$ and μ_0, hence also find radius of curvature of the beam near the origin. Express your answer in terms of μ_0, $\mu(y)$ and $d\mu/dy$.

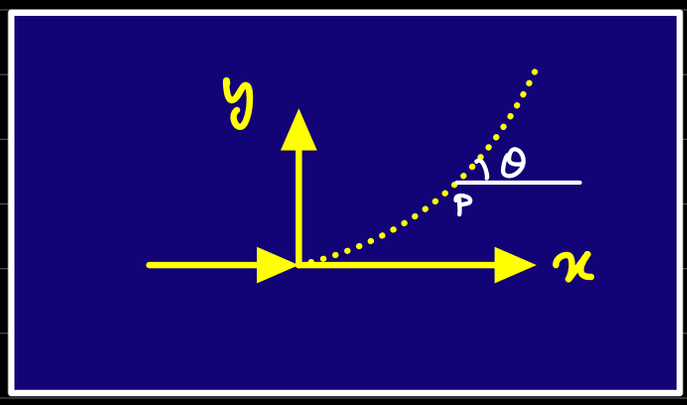


Solution; Consider a point P on the path where the curve is inclined at an angle θ as shown.

Now by snell's law:

$$\mu_0 \sin 90° = \mu\cos\theta$$

$$\Rightarrow \cos\theta = \frac{\mu_0}{\mu}$$



$$\Rightarrow \tan\theta = \frac{dy}{dx} = \frac{\sqrt{\mu^2 - \mu_0^2}}{\mu_0} \quad ①$$

$$R = \frac{(1+y'^2)^{3/2}}{y''} = \left(\frac{\mu}{\mu_0}\right)^3 \cdot \frac{1}{y''} \text{ Ans.}$$

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Now differentiating ① wrt x :

$$\frac{d^2y}{dx^2} = \frac{1}{\mu_0}\left[\frac{1}{2\sqrt{\mu^2-\mu_0^2}} \cdot 2\mu\frac{d\mu}{dx}\right] = \frac{1}{\mu_0}\frac{\mu}{\sqrt{\mu^2-\mu_0^2}} \cdot \frac{d\mu}{dy} \times \frac{dy}{dx} \quad ②$$

Now using ① in ② : $y'' = \frac{\mu}{\mu_0\sqrt{\mu^2-\mu_0^2}} \cdot \frac{\sqrt{\mu^2-\mu_0^2}}{\mu_0} \cdot \frac{d\mu}{dy}$

$$\Rightarrow y'' = \frac{\mu}{\mu_0^2} \cdot \frac{d\mu}{dy}$$

Now at the origin, $R = \frac{1}{y''} = \frac{\mu_0^2}{\mu\,(d\mu/dy)} = \frac{\mu_0}{(d\mu/dy)}$ Ans.

(∵ at origin $\mu = \mu_0$). Thus $\boxed{R = \mu_0/(d\mu/dy)}$

EXAMPLE: PATHFINDER: CHECK 8:

Refractive index μ of atmosphere around a planet varies with altitude h according to equation $\mu = \mu_0 - kh$. Here μ_0 and k are positive constants. Denoting radius of the planet by R, find the altitude at which a point flash of light spreads around the planet in a spherical layer concentric with the planet in addition to other directions.

Solution: Here if the flash is spreading like a concentric sphere around planet then for a horizontally traveling beam, the radius of curvature must be equal to $(R+h)$, where h is the altitude at which this happens. Now if vertical direction is chosen as y direction then, $\frac{d\mu}{dy} = \frac{d\mu}{dh} = -k$, thus radius of curvature of the horizontal beam (as derived in previous example) : $R = \mu_0/(d\mu/dy)$

$\Rightarrow R = (\mu_0 - kh)/(-k)$, negative sign only signified concave down.

Thus $|R| = \frac{1}{k}(\mu_0 - kh) = (R+h) \Rightarrow h = \frac{1}{2}\left(\frac{\mu_0}{k} - R\right)$ Ans.

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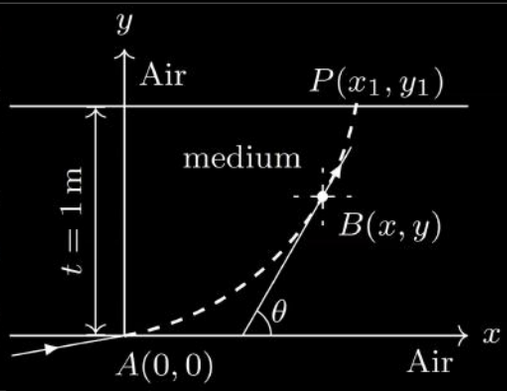
EXAMPLE: JEE 1995:

JEE: A ray of light travelling in air is incident at grazing angle (incident angle $= 90°$) on a long rectangular slab of a transparent medium of thickness $t = 1.0\,\text{m}$. The point of incidence is the origin $A(0,0)$. The medium has a variable index of refraction $n(y)$ given by $n(y) = \left[ky^{3/2} + 1\right]^{1/2}$, where $k = 1.0\,\text{m}^{-3/2}$. The refractive index of air is 1.0. (a) Find trajectory equation (b) Find coordinates of the exiting point.



Solution: This is a parallel slab like geometry with continuous variation of refractive index. Thus applying snell's law at a general point:

$$1 \sin 90° = \mu \cos\theta \Rightarrow \cos\theta = 1/\mu \Rightarrow \frac{dy}{dx} = \tan\theta = \sqrt{\mu^2 - 1}$$

$$\text{Thus } dy/dx = k\,y^{3/4} \Rightarrow \int_0^y y^{-3/4}\,dy = \int_0^x k\,dx \Rightarrow 4y^{1/4} = kx$$

Thus required trajecton:

$$y = (kx/4)^4$$

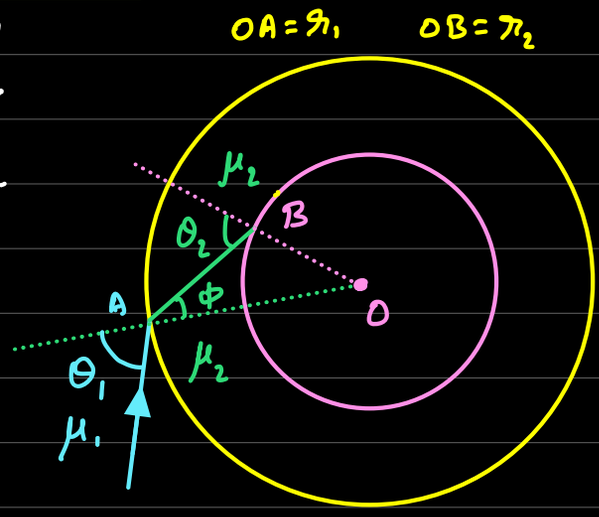
Now at exit, $y = 1 \Rightarrow 1 = (x/4)^4 \Rightarrow x = 4$ ans.

(Since final medium is air again ray will come out nearly parallel to the surface. (Actually parallel to incident ray).

SNELL'S LAW FOR SPHERICALLY SYMMETRIC MEDIA:

Consider a ray incident through medium of refractive index μ_1 on to a medium of refractive index μ_2, as shown in the figure.

Now, applying Snell's law :



$$\mu_1 \sin\theta_1 = \mu_2 \sin\phi \quad (1)$$

Now consider ΔOAB:

$$\frac{OB}{\sin\phi} = \frac{OA}{\sin\theta_2} \Rightarrow \sin\phi = \frac{OB}{OA}\sin\theta_2$$

$$\Rightarrow \sin\phi = \frac{r_2}{r_1}\sin\theta_2 \quad (2) \quad \text{using (2) in (1):}$$

$$\mu_1 \sin\theta_1 = \mu_2 \times (r_2/r_1)\sin\theta_2 \Rightarrow \boxed{\mu_1 r_1 \sin\theta_1 = \mu_2 r_2 \sin\theta_2}$$

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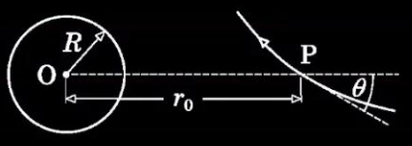
Extending this to several concentric media (or continuously varying spherically symmetric media):

$$\mu r \sin\theta = \text{Constant}$$

Example: Pathfinder: Challenge 2:



A laser beam propagates through a spherically symmetric medium surrounding a metal sphere of radius $R = 10$ cm. Refractive index of the medium varies with distance r from centre O of the sphere according to the law $\mu(r) \propto r$. Here $R \ll r < \infty$.

The laser beam makes angle of $\theta = 30°$ with a radial line at point P, which is $r_0 = 50\sqrt{2}$ cm away from O. What minimum distance from surface of the sphere can the beam reach?

Solution: This is a direct application of Snell's law for spherically symmetric media. At minimum distance, the angle of incidence is 90° (∵ radial component of velocity is zero). Now let $\mu(r) = kr$.

Thus:

$$k \times r_0 \sin\theta = k' r_{min} \sin 90°, \text{ Now } k \propto r \Rightarrow r_0^2 \sin\theta = r_{min}^2$$

$$\Rightarrow r_{min} = r_0\sqrt{\sin\theta} = 50\sqrt{2} \times \frac{1}{\sqrt{2}} = 50 \text{ cm}, \text{ thus min distance from surface: } 50 - 10 = 40 \text{ cm Ans.}$$

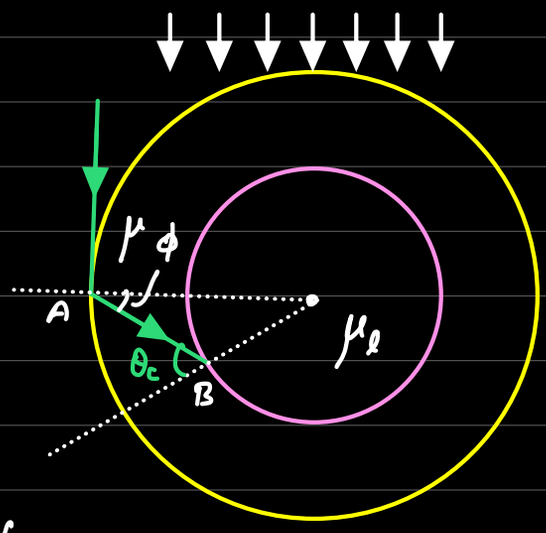
Example: Pathfinder: Build: 17:

A capillary of outer radius r_0 made of a glass of refractive index μ is filled with a liquid of refractive index μ_l ($\mu_l < \mu$). What should the minimum internal radius of the capillary be so that all the light incident on the capillary enter the liquid? (Ignore partial Reflections).

Solution: For the extreme incident ray to enter the liquid, its incidence angle can't exceed the critical angle. Now applying Snell's law for spherically symmetric media in limiting case:

$$1 r_0 \sin 90° = \mu r \sin\theta_c \text{ where } r \text{ is the inner radius and } \sin\theta_c = \mu_l/\mu, \text{ thus:}$$

$$r_0 = \mu_l r \Rightarrow r = r_0/\mu_l \text{ Ans.}$$



(Here there is cylindrical symmetry instead of spherical symmetry, but mathematics of it essentially remains the same).

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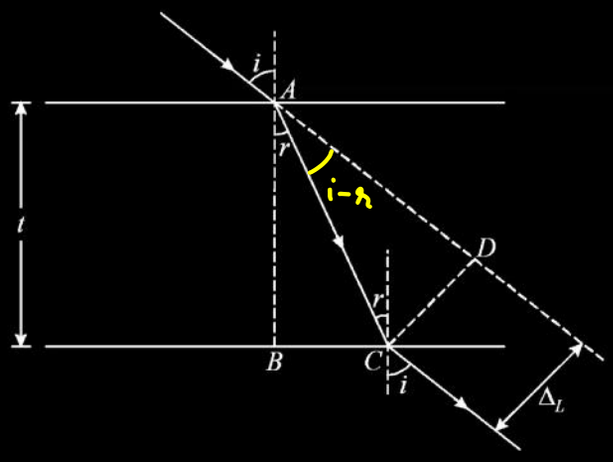
LATERAL SHIFT FROM A GLASS SLAB:



Recall that with symmetric medic on both sides of the slab, the emergent ray is parallel to the incident ray. The perpendicular distance between these lines is called lateral shift (See fig.).

Now consider ΔABC, from this, $AC = t\sec r$.

Now in ΔACD, $CD = AC\sin(i-r)$

$\Rightarrow \Delta_L = t\sec r \sin(i-r)$

$\Rightarrow$ Lateral shift: $\boxed{\Delta_L = t\sin(i-r)/\cos r}$

If i and r are small angles then we can simplify further,

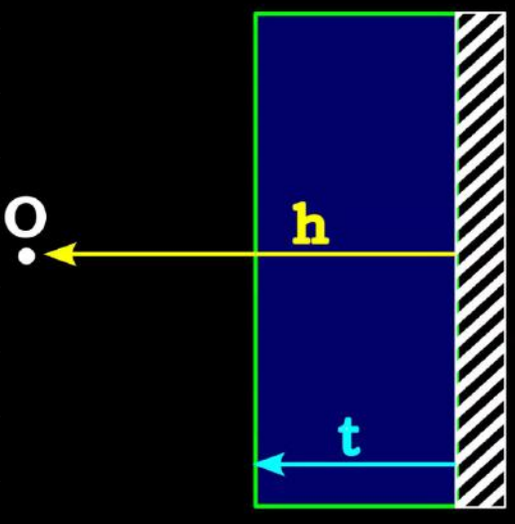
$\sin i = i$ and $\sin r = r \approx \sin i/\mu \approx i/\mu$, $\cos r \approx 1$

thus,

$$\boxed{\Delta_L = ti(1-1/\mu)} = \text{Normal shift } \times i.$$

($\because$ Normal shift $x = t(1-1/\mu)$)

IMAGE FORMATION FROM THICK PLANE MIRROR (SILVERED SLAB)



Imagine an object O at a distance h from silvered surface of a thick mirror. To find its image, we can use one of the following methods:

METHOD I: Imagining a thin layer of air between silvered surface and glass, we can treat the system as a combination of a slab and a mirror. Thus due to first refraction from slab, the object appears $t(1-\frac{1}{\mu})$ distance closer to the mirror.

Thus due to reflection, image is formed a distance $h - t(1-\frac{1}{\mu})$ behind the mirror. Now the reflected light again passes through the slab thus image is further brought nearer by a distance $t(1-\frac{1}{\mu})$ thus

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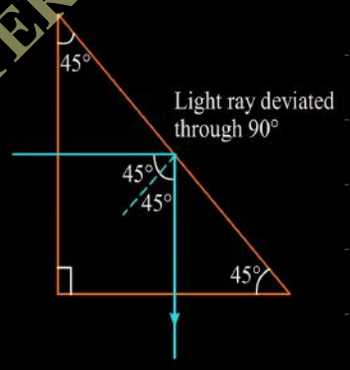
final image is formed $h - 2t\left(1 - \frac{1}{\mu}\right)$ distance behind the mirror.

METHOD II: Here we consider apparent distance of the mirror. The Apparent distance of the mirror from object is $h - t\left(1 - \frac{1}{\mu}\right)$. Thus image is formed at a distance $2\left[h - t\left(1 - \frac{1}{\mu}\right)\right]$ from the object. Thus distance from mirror is $2\left[h - t\left(1 - \frac{1}{\mu}\right)\right] - h$ i.e. $h - 2t\left(1 - \frac{1}{\mu}\right)$. Thus we get the same result from this technique also.

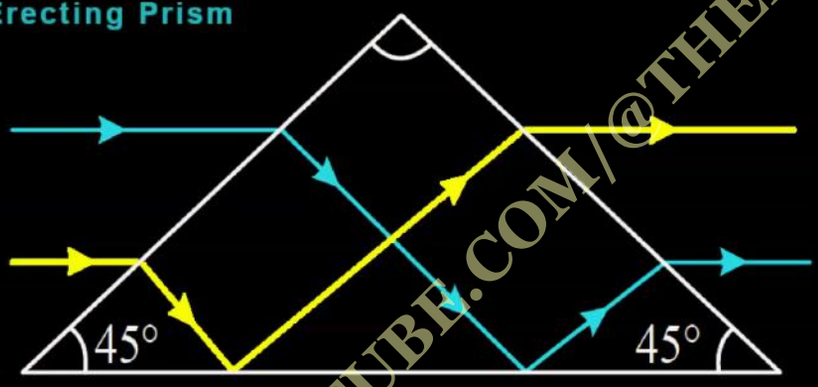
TIR FROM ISOSCELES RIGHT ANGLED PRISMS:

Critical angle for glass is about 42°. Thus by suitably choosing angle of incidence on isoceles right angled prism, we can use it to bend a ray through 90°, 180° and also use it for erecting an inverted image as illustrated in adjacent and following figures.

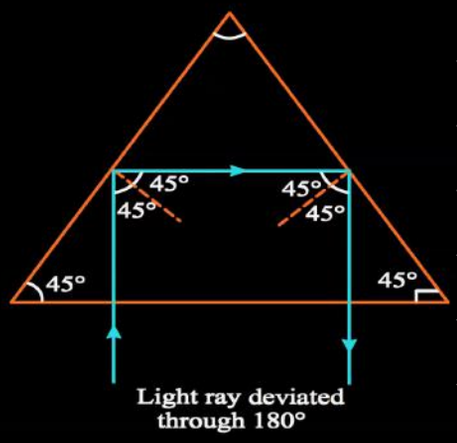
Deviation through 90^0:



Erecting Prism



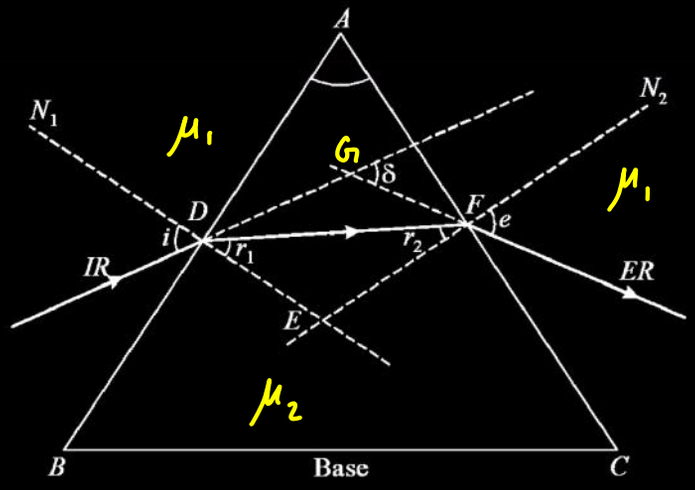
Deviation through 180^0:



REFRACTION FROM PRISMS:

The adjacent figure shows refraction from a prism when TIR is not happening at any surface. Here ∵ $\angle ADE = \angle AFE = 90°$, ADEF is a cyclic quadrilateral, thus for this, $\angle A + \angle E = 180°$, also in ΔDEF, $r_1 + r_2 + \angle E = 180°$, thus we have our first prism relation:

$$r_1 + r_2 = A$$



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→ Note that in the equation above r_1 and r_2 are signed values they are considered positive if the refracted ray is above the normal in the media (for second surface, incident ray should be above the normal for r_2 to be positive).

→ Similarly i, can be negative if incidence is from above the Normal.

ANGLE OF DEVIATION: The angle between the incident ray and the final emergent ray is called the angle of deviation. From ΔGDF,

$\delta = \angle D + \angle F$ i.e. $i - r_1 + e - r_2 = i + e - (r_1 + r_2)$ thus,

$$\boxed{\delta = i + e - A}$$

In terms of r_1, $i = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\sin r_1\right)$ and $e = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\sin r_2\right)$

Thus $\delta = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\sin r_1\right) + \sin^{-1}\left[\frac{\mu_2}{\mu_1}\sin(A - r_1)\right] - A$

Now δ is simply expressed as a function of r_1 and has a form

$\delta = f(r_1) + f(A - r_1)$ where f is an appropriate function.

Now for minimum deviation, $d\delta/dr_1 = 0$

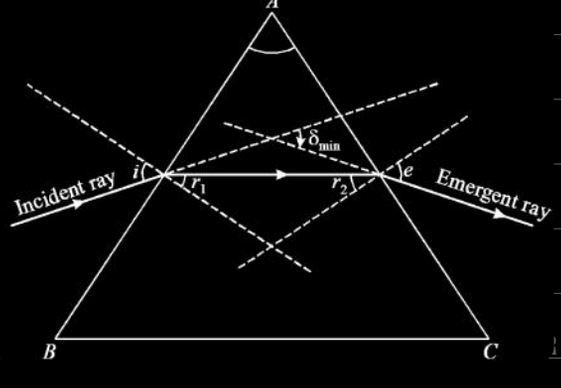
$\Rightarrow$

$f'(r_1) - f'(A - r_1) = 0$ One obvious root of this equation is $r_1 = A - r_1$ or $r_1 = A/2$, at this $r_2 = A - r_1 = A/2$, thus at minimum deviation:

$$\boxed{r_1 = r_2 = A/2}$$

→ This also means that for an isosceles prism with incidences on equal sides, the ray inside the prism is parallel to the base in case of minimum deviation.

→ At minimum deviation the figure has a symmetry as visible from adjacent figure, there $i = e$.



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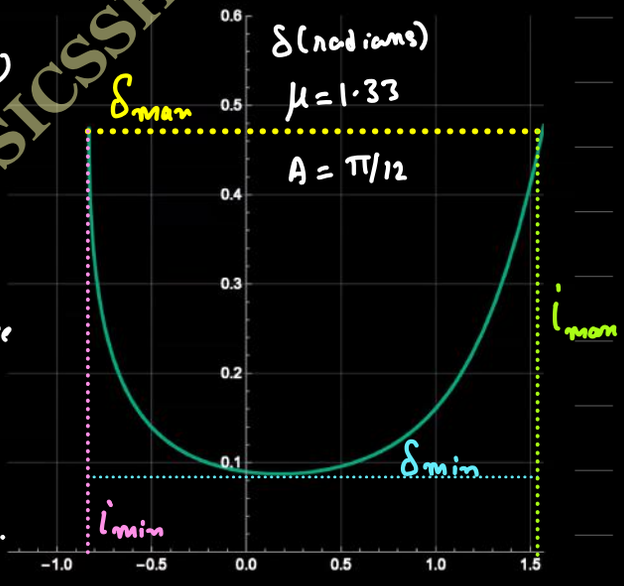
Relation Between Refractive Index And Minimum Angle Of Deviation:

Recall that angle of deviation from a prism is given by:
$\delta = i + e - A$, now at minimum deviation, $i = e$ thus,
$\delta_{min} = 2i - A$ or $i = (A + \delta_{min})/2$, also $r_1 = r_2 = r = A/2$.
Now using Snell's law at incidence surface, (Assuming $\mu_1 = 1, \mu_2 = \mu$)

$$1 \sin\left[(A+\delta_m)/2\right] = \mu \sin(A/2) \Rightarrow \boxed{\mu = \frac{\sin\left[(A+\delta_m)/2\right]}{\sin(A/2)}}$$

Plot Of Angle Of Deviation Vs Incidence Angle:

The adjacent figure shows the graph of angle of deviation of a water prism ($\mu = 1.33$) and refracting angle 15°. From this graph we can infer that there is certain minimum angle of incidence for ray to emerge out and maximum angle of incidence is $\pi/2$. For $i = i_{min}$, $e = \pi/2$ and for $i = \pi/2$, $e = i_{min}$, this can be inferred from principle of reversibility. (If the path of the ray is reversed, i and e get swapped while δ remains the same).



If the incidence angle is less than i_{min} there is TIR on second surface.

→ Maximum deviation corresponds to $i = i_{min}$ and $i = \pi/2$, this also corresponds to ray exiting at grazing emergence (or borderline TIR) and grazing incidence (entry at critical angle respectively).

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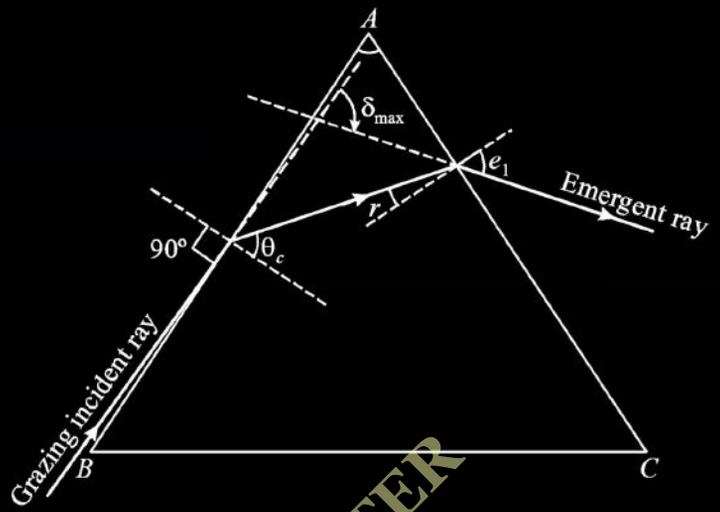
Maximum Angle Of Deviation And Minimum Angle Of Incidence:

Recall that maximum deviation occurs for $i = \pi/2$ and $i = i_{min}$ which corresponds to e for $i = \pi/2$ (by principle of reversibility). Now consider the adjacent figure,



Now with this figure, applying Snell's law on second refracting surface,

$$\mu \sin(A - \theta_c) = 1 \sin e$$

$$\Rightarrow \sin e = \mu [\sin A \cos\theta_c - \cos A \sin\theta_c], \text{ But } \sin\theta_c = 1/\mu \text{ thus:}$$

$$\sin e = \mu \left[\sin A \sqrt{1 - 1/\mu^2} - \cos A \times 1/\mu \right]$$

$$\Rightarrow \sin e = \sqrt{\mu^2 - 1}\, \sin A - \cos A, \text{ Here } e \text{ is minimum possible emergence angle also (} \because \text{ when } r_1 \text{ is max, } r_2 \text{ is min so also } e).$$

Thus $$\boxed{e_{min} = i_{min} = \sin^{-1}\left[\sqrt{\mu^2 - 1}\, \sin A - \cos A \right]}$$

Now δ_{max} is simply $\pi/2 + e_{min} - A$, thus:

$$\boxed{\delta_{max} = \pi/2 + e_{min} - A}$$

→ If the argument of $\sin^{-1}$ function becomes more than 1, the e_{min} is not defined which means that TIR will happen on second surface no matter what the angle of incidence. Note that for prism in Air the argument of $\sin^{-1}$ function can never become less than -1.

→ Thus for no emergence from second refracting surface:

$$\boxed{\sqrt{\mu^2 - 1}\, \sin A - \cos A > 1}$$

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$$\Rightarrow \quad (\mu^2-1)\sin^2 A > (1+\cos A)^2 = 1+\cos^2 A + 2\cos A$$

$$\Rightarrow \quad \mu^2 \sin^2 A > 2 + 2\cos A = 4\cos^2(A/2)$$

$$\Rightarrow \quad 2\mu \sin\frac{A}{2}\cos A/2 > 2\cos A/2 \Rightarrow \sin(A/2) > \frac{1}{\mu} \Rightarrow A > 2\sin^{-1}\frac{1}{\mu} = 2\theta_c$$

Thus for no emergence from other surface irrespective of incidence angle,

$$\boxed{A > 2\theta_c}$$

→ If the surrounding medium were some liquid instead of air, we will have to replace $\mu \longrightarrow \mu/\mu_l$ in all the above formulae where μ_l is refractive index of liquid.

→ If $\mu_l > \mu$ then any ray which enters the prism will exit (since prism is rarer medium now) however now i will be limited in the range $i \in (-\theta_c, \theta_c)$, similarly e will also have a limited range $e \in (-\theta_c, \theta_c)$, or even stricter range depending on geometry.

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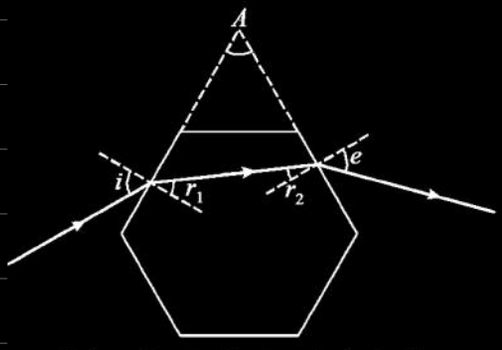
EXAMPLE: Some text is written on a paper and a glass cube is placed over it. Find the minimum refractive index of the cube so that the text is not visible from its vertical faces.

Solution: The right angle formed at lower right edge can be considered to be the refracting angle of right angled prism. Thus the problem essentially reduces to the case where there is no emergence irrespective of angle of incidence on the prism. For this we proved that $A > 2\theta_c$ thus:

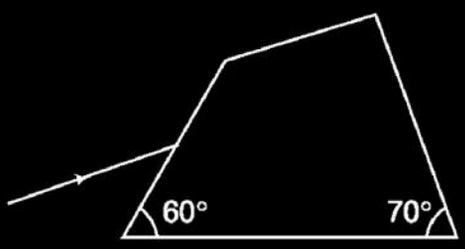
$$90^\circ > 2\theta_c \Rightarrow \theta_c < 45^\circ \Rightarrow \frac{1}{\mu} < \sin 45^\circ = \frac{1}{\sqrt{2}} \Rightarrow \mu > \sqrt{2}$$

Thus our required answer: $\mu > \sqrt{2}$ ans.

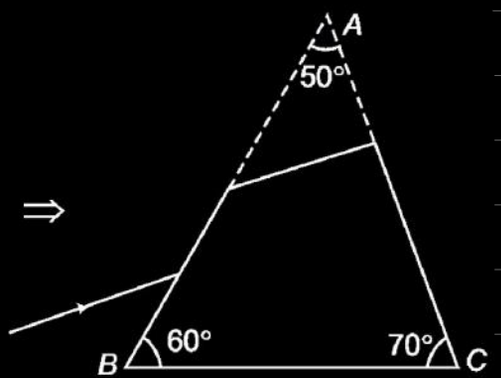
Visualizing Complete Prism:



Light refraction by hexahedral prism (cross-sectional view)



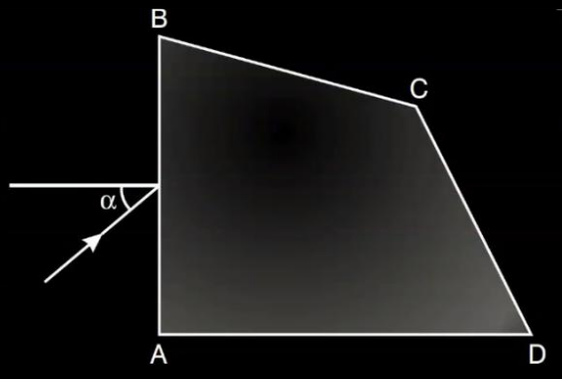
(a)

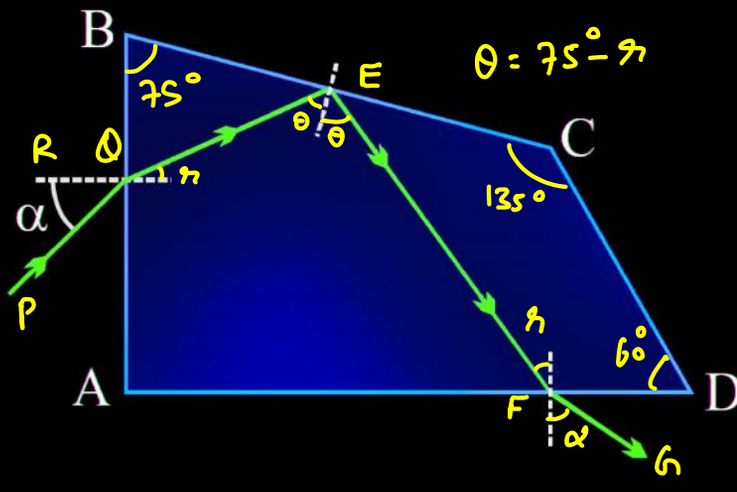


(b)

Sometimes, instead of a triangular prism we may be given a hexagonal prism, a quadrilateral prism or some other polygonal prism. In such cases, many times it is helpful to visualize an equivalent triangular prism by extrapolating the sides as shown in figures above.

EXAMPLE The faces of prism $ABCD$ made of glass of refractive index μ form dihedral angles $\angle A = 90°$, $\angle B = 75°$, $\angle C = 135°$ and $\angle D = 60°$. A beam of light falls on face AB and after complete internal reflection from face BC escapes through face AD. Find the range of μ and angle of incidence α of the beam, if a beam that has passed through the prism in this manner, is perpendicular to the incident beam.





Solution: $PQ \perp FG$ thus $\angle DFG = 90° - \alpha$ (1)
Thus refraction angle at Q must be equal to incidence angle at $F = r$ (say).
Also TIR is happening at E so let the incidence and reflection angles there both be $\theta = 75° - r$. Now consider Quad ECDF:

$$\underset{E}{[90° - (75° - r)]} + \underset{C}{135°} + \underset{D}{60°} + \underset{F}{(90° + r)} = 360°$$

Thus $r = 30°$, $\theta = 45°$. Now $\theta > \theta_c = \sin^{-1}(1/\mu)$, Also $r < \theta_c \Rightarrow \mu < 2$

$\Rightarrow 45° > \sin^{-1}(1/\mu) \Rightarrow \frac{1}{\mu} < \frac{1}{\sqrt{2}}$ or $\mu > \sqrt{2}$ ans

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Now $1 \sin\alpha = \mu \sin 30^\circ$ (Snell's law at Q)

$\Rightarrow \alpha = \sin^{-1}(\mu \sin 30^\circ) = \sin^{-1}(\mu/2)$, also, $\mu > \sqrt{2}$

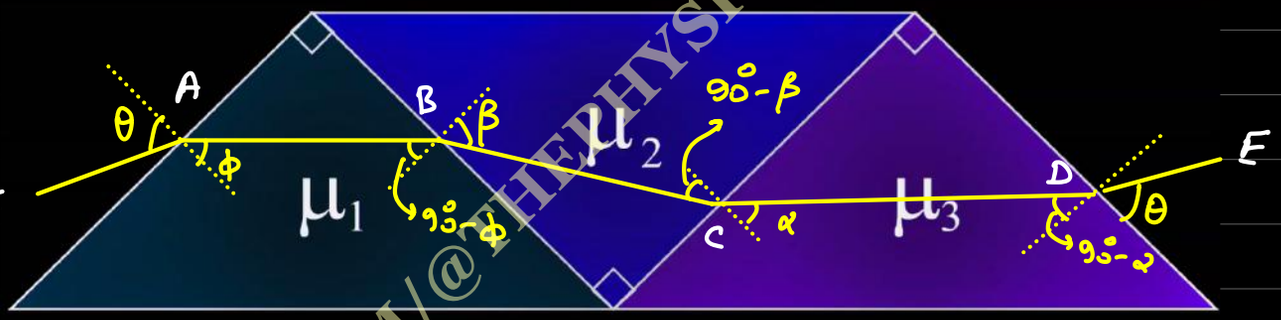
$\Rightarrow \alpha > 45^\circ$ Ans. (For a unique value of $\mu > \sqrt{2}$, α will be unique).

Thus $\mu \in (\sqrt{2}, 2)$, $\alpha \in (45^\circ, 90^\circ)$

EXAMPLE:

Three right angled prisms of refractive indices μ_1, μ_2 and μ_3 are joined together so that the faces of the middle prism in are in contact each with one of the outside prisms. If the ray passes through the composite block undeviated, show that $\mu_1^2 + \mu_3^2 - \mu_2^2 = 1$.

Solution:



For incident and final rays to be parallel, a tentative ray diagram is shown above. Now, using Snell's law at various points:

$1 \sin\theta = \mu_1 \sin\phi$ (1) (A)

$\mu_2 \sin\beta = \mu_1 \cos\phi$ (2) (B)

$\mu_2 \cos\beta = \mu_3 \sin\alpha$ (3) (C)

$1 \cos\theta = \mu_3 \cos\alpha$ (4) (D)

Now squaring and adding all the equations:

$1 + \mu_2^2 = \mu_1^2 + \mu_3^2 \Rightarrow \mu_1^2 + \mu_3^2 - \mu_2^2 = 1$ QED.

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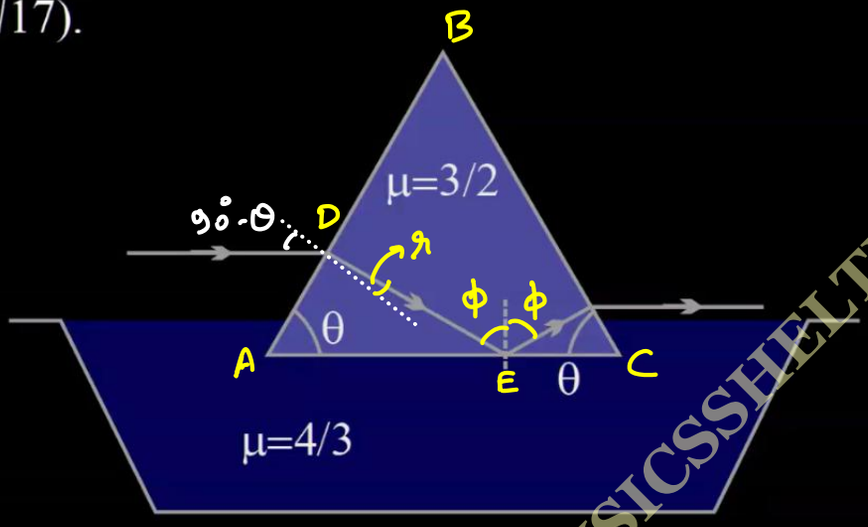
EXAMPLE:

An isosceles triangular glass prism stands with its base in water as shown. The angles that its two equal sides make with the base are θ each. An incident ray of light parallel to the water surface internally reflects at the glass-water interface and subsequently re-emerges into the air. Taking the refractive indices of glass and water to be 3/2 and 4/3 respectively, show that θ must be at least $\tan^{-1}(2/\sqrt{17})$.



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Solution: Consider ΔADE as a prism with refractive angle at A. It is then a case of negative incidence angle and negative r. Thus:

$\theta = \phi - r$ (1) In the limiting case, $\phi = \theta_c \Rightarrow \phi = \sin^{-1}(8/9)$ (2)

Also using snell's law at D: $1\cos\theta = \frac{3}{2}\sin r$ (3)

$\Rightarrow 1\cos\theta = \frac{3}{2}\sin[\phi - \theta]$

$\Rightarrow 1\cos\theta = \frac{3}{2}[\sin\phi\cos\theta - \cos\phi\sin\theta] \Rightarrow \cos\theta[\frac{3}{2}\sin\phi - 1] = \frac{3}{2}\sin\theta\cos\phi$

$\Rightarrow \tan\theta = \frac{2}{3\cos\phi}[\frac{3}{2}\sin\phi - 1]$ (4) $= \frac{2}{3\cos\phi}[\frac{3}{2}\times\frac{8}{9} - 1] = \frac{2}{9\cos\phi}$

Now $\cos\phi = \frac{\sqrt{81-64}}{9} = \frac{\sqrt{17}}{9}$, thus $\tan\theta = 2/\sqrt{17}$

Thus required θ is given by $\tan\theta > 2/\sqrt{17}$

(We infer the greater than sign from equation (4), upon increasing ϕ, θ will also increase).

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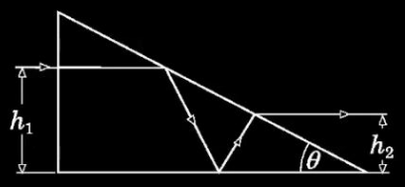
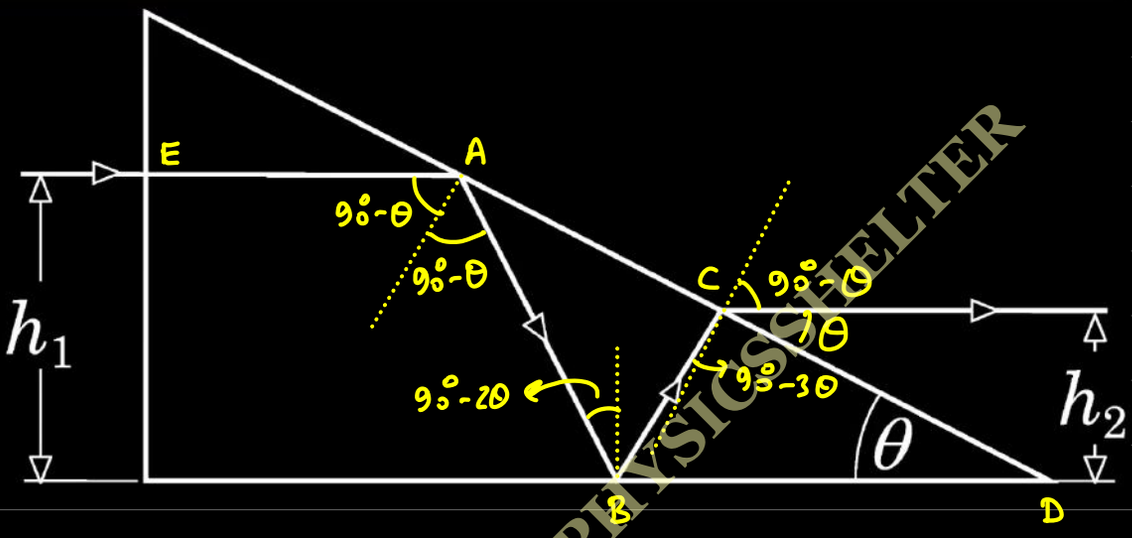
Example: Pathfinder: MCQ 12–14:



Consider the setup shown in the figure, which uses a right-angled prism to transmit a horizontal incident ray. This device is known as anamorphic de-magnifier. During its passage through the prism, the ray experiences total internal reflection (TIR) twice and then exits as shown in the figure. The prism is made from a glass of refractive index μ.

Find (a) $\mu(\theta)$ (b) Range of θ in terms of μ (c) Ratio h_1/h_2. (in θ).

Solution:



Consider the diagram above, Applying snell's law at C:

$$\mu \sin(90°-3\theta) = 1\cos\theta \Rightarrow \mu = \cos\theta/\cos(3\theta) \quad \text{Ans.}$$

Now TIR should occur at B and not occur at C thus:

$$90°-2\theta > \sin^{-1}(1/\mu) \quad \text{and} \quad (90°-3\theta) < \sin^{-1}(1/\mu)$$

$$\Rightarrow \frac{1}{\mu} < \cos 2\theta \quad \text{and} \quad \frac{1}{\mu} > \cos 3\theta \quad \text{Now cos is a decreasing fn.}$$

Thus: $$\theta < \frac{1}{2}\cos^{-1}\left(\frac{1}{\mu}\right) \quad \text{and} \quad \theta > \frac{1}{3}\cos^{-1}\left(\frac{1}{\mu}\right)$$

i.e $$\frac{1}{3}\cos^{-1}(1/\mu) < \theta < \frac{1}{2}\cos^{-1}(1/\mu) \quad \text{Ans.}$$

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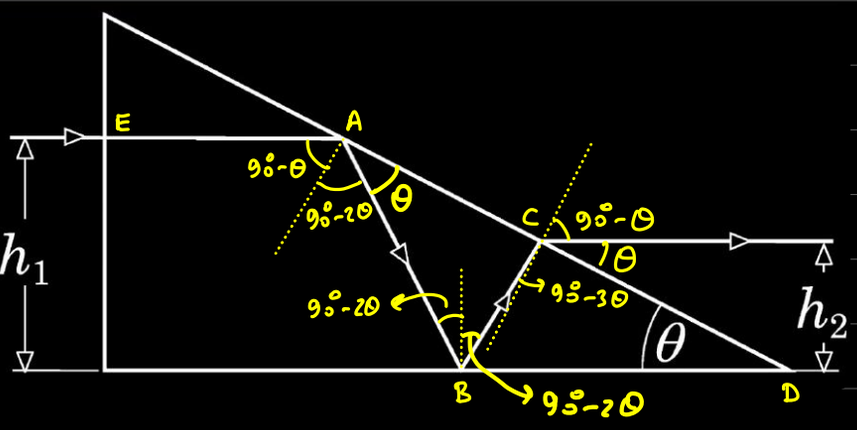
Now both AB and BC are inclined at $90^\circ - 2\theta$ with vertical (see vertex B). Thus:

$$AB = \frac{h_1}{\cos(90^\circ - 2\theta)} = \frac{h_1}{\sin 2\theta} \quad (1)$$

Similarly $BC = \frac{h_2}{\cos(90^\circ - 2\theta)} = \frac{h_2}{\sin 2\theta}$ (2)



Also: $\frac{BC}{\sin\theta} = \frac{AB}{\sin 3\theta}$ ($\because \angle C = 3\theta$ in ΔABC).

$$\Rightarrow \frac{h_2}{\sin 2\theta \cdot \sin\theta} = \frac{h_1}{\sin 2\theta \cdot \sin 3\theta} \Rightarrow \frac{h_1}{h_2} = \frac{\sin 3\theta}{\sin\theta} \text{ Ans.}$$

→ If we wish we can also express h_1/h_2 in terms of μ.

$$\frac{h_1}{h_2} = \frac{\sin 3\theta}{\sin\theta} = \frac{3\sin\theta - 4\sin^3\theta}{\sin\theta} = 3 - 4\sin^2\theta = -1 + 4\cos^2\theta \quad (4)$$

Also from part (a) $\mu = \frac{\cos\theta}{\cos 3\theta} = \frac{\cos\theta}{4\cos^3\theta - 3\cos\theta} = \frac{1}{4\cos^2\theta - 3}$ (5)

Now eliminating $\cos^2\theta$ from (4) & (1) we get:

$$\frac{h_1}{h_2} = (1+2\mu)/\mu \text{ Ans.}$$

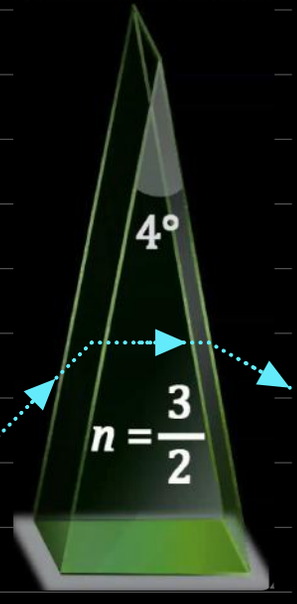
Refraction From Thin Prisms At Near Normal Incidence:

Recall that angle of deviation from a prism is given by $\delta = i + e - A$. Now if A is very small and i is very small then r_1 and r_2 are also very small, Now $\sin i = \mu \sin r_1 \Rightarrow i \approx \mu r_1$ similarly $e \approx \mu r_2$ thus:

$$\delta = \mu r_1 + \mu r_2 - A$$

$$\Rightarrow \delta = \mu(r_1 + r_2) - A = A(\mu - 1)$$

Thus $\boxed{\delta = (\mu - 1)A}$



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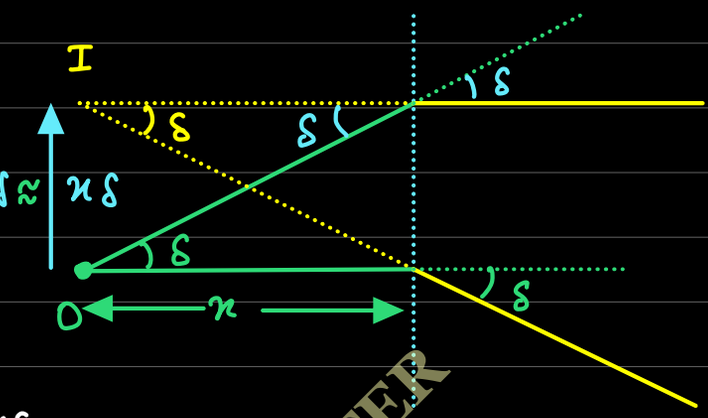
→ For thin prisms, angle of deviation is independent of incidence angle so long as incidence angle is small.

Image Formation From Thin Prisms: (Near Normal Incidence):

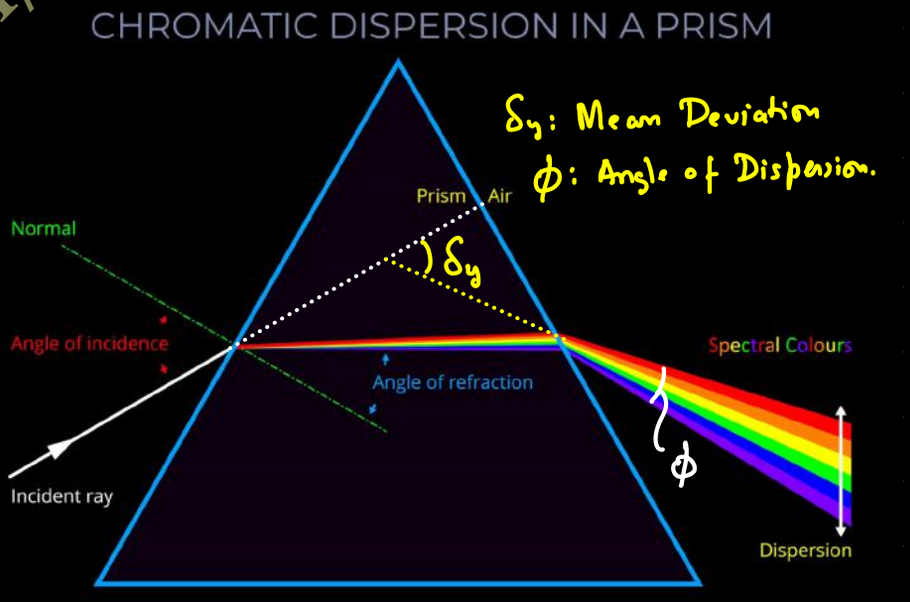


Since the prism is very thin, for sake of convenience we can represent it just by a straight line which deflects any incident ray through an angle δ.
We have chosen two convenient rays for forming image of our object and as clear from the diagram, the image is formed vertically above the object (i.e shifted parallel to the prism line) at a distance $h = x\delta = x(\mu-1)A$.
Thus:

$$\boxed{h = x(\mu-1)A}$$

h: distance of image from object
x: distance of object from prism.

Dispersion And Deviation From A Prism:



δ_y: Mean Deviation
ϕ: Angle of Dispersion.

Whenever a white ray is incident on a prism, due to varying refractive indices for various wavelengths, the beam is split in to Rainbow colors with violet at the bottom and red at the top.
(∵ μ_v is highest and μ_r is lowest).

Angle Of Mean Deviation: The angle between the emergent Yellow ray and the incident white ray is called angle of mean deviation (δ_y) because yellow light wavelength is almost at the mid point of the visible spectrum (400 - 700 nm). 550 nm is a green-ish Yellow wavelength which is exact mid point of visible

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spectrum.

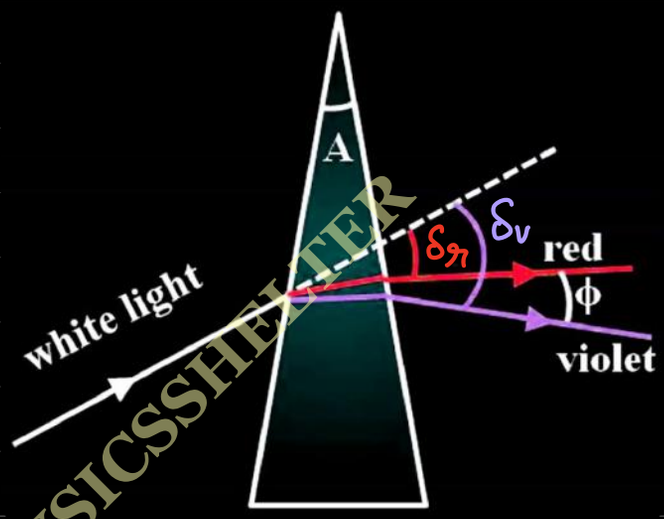
ANGLE OF DISPERSION: The total angle in to which the emergent beam spreads is called the angle of dispersion (ϕ), (See figure). This can also be defined as the angle between the highest emergent ray (red) and lowest emergent ray (violet).

DISPERSIVE POWER OF A MATERIAL:

The ratio of angle of dispersion (ϕ) to the angle of mean deviation (δ_y) as obtained for a thin prism of a given material is called the dispersive power of that material.

Mathematically:

$$\omega = \phi / \delta_y$$



ω: Dispersive power. Now if δ_v and δ_r are deviations for violet and red beams, then:

$\phi = \delta_v - \delta_r$ thus another definition of ω can be given as:

$$\omega = (\delta_v - \delta_r) / \delta_y$$

Now for thin prisms; $\delta = (\mu - 1)A$ as seen in a previous section.

Thus,

$$\omega = \frac{(\mu_v - 1)A - (\mu_r - 1)A}{(\mu_y - 1)A}$$

, μ_v, μ_r and μ_y are refractive indices for violet, red and yellow respectively.

Thus $$\omega = (\mu_v - \mu_r) / (\mu_y - 1)$$

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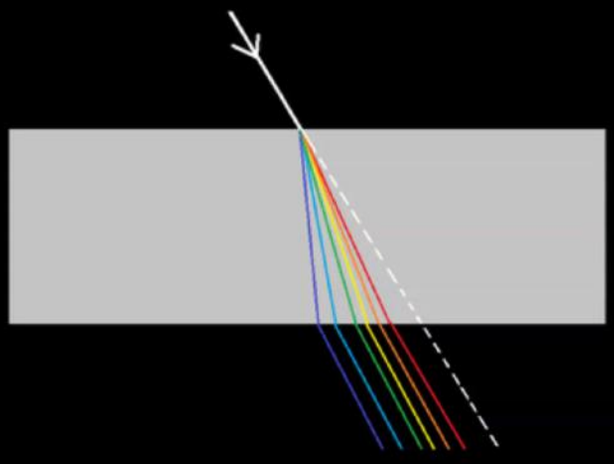
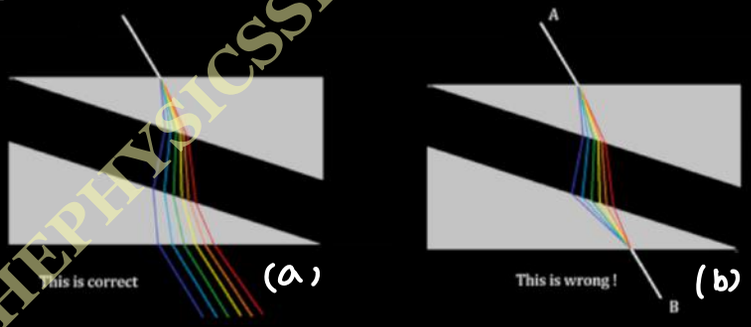
DISPERSION FROM A SLAB:



As we have studied in a previous section, any ray incident on a glass slab emerges parallel to itself, thus every white ray is dispersed in to its component wavelengths. However we normally do not observe this practically. This is because we can't isolate a mathematical ray. We always have a bundle of rays and because of this violet part of one ray will become collinear with red part of another ray from the bundle (and similarly for all other colors), thus we see the emergent beam as white. Sometimes textbooks and internet represent the slab a a combination of two prisms but present a wrong ray diagram (see figure b). The correct ray diagram is as shown in figure (a)



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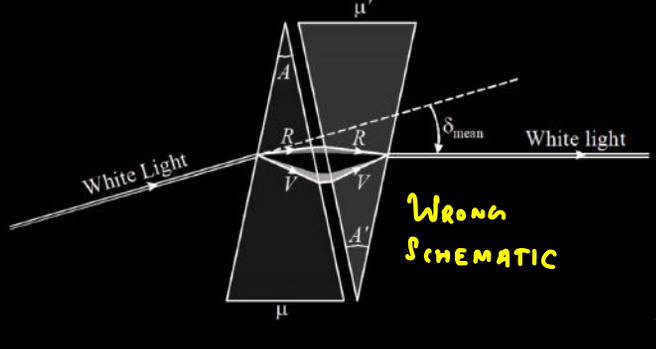
ACHROMATIC COMBINATION OF PRISMS: DEVIATION WITHOUT DISPERSION

Just as a glass slab (modeled as two prisms with oppositely directed refracting angles) can have an almost white emergent beam for a white incident beam, similarly if two prisms are combined in such a way that combined deviation from two prisms for violet color is equal to that of red color then final red beam will be parallel to the violet beam and thus we will have a nearly white emergent beam.

Now if two prisms have refracting angles A and A' and refractive indices μ_v and μ'_v and μ_r and μ'_r Then from the combination:

$$\delta_{v_{net}} = \delta_v + \delta'_v = (\mu_v - 1)A - (\mu'_v - 1)A'$$



Similarly $\delta_{r_{net}} = \delta_r + \delta_r' = (\mu_r - 1)A - (\mu_r' - 1)A'$, now for deviation without dispersion, $\delta_{v_{net}} = \delta_{r_{net}}$

$$\Rightarrow (\mu_v - 1)A - (\mu_v' - 1)A' = (\mu_r - 1)A - (\mu_r' - 1)A'$$

$$\Rightarrow \boxed{(\mu_v - \mu_r)A = (\mu_v' - \mu_r')A'} \quad (1)$$

→ With this mean deviation produced is $\boxed{\delta_y = (\mu_y - 1)A - (\mu_y' - 1)A'}$ (2)

Now we can also express (1) in terms of dispersive powers:

$$\frac{(\mu_v - \mu_r)A}{(\mu - 1)} \times (\mu - 1) = \frac{(\mu_v' - \mu_r')A}{(\mu' - 1)} (\mu' - 1) \quad (\mu \& \mu' \text{ are for Yellow})$$

Thus for achromatic combination:

$$\boxed{(\mu - 1)\omega A = (\mu' - 1)\omega' A'}$$

→ This formula can be extended to an achromatic combination of n prisms also as:

$$\boxed{\Sigma (\mu_i - 1)\omega_i \cdot A_i = 0}$$

Here A_i are signed, for inverted prisms, A_i is considered negative.

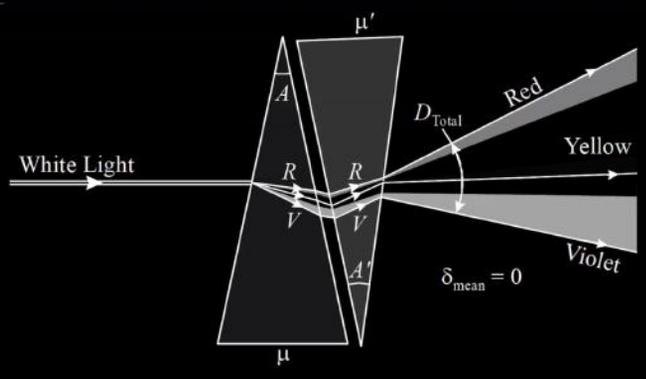
Prism Combination for Dispersion Without Mean Deviation (Direct Vision):

If through a combination of prisms, net deviation of the mean wavelength (Yellow) is zero, we call it direct vision combination (or Dispersion without deviation case). Thus for this case,

$\delta_y - \delta_y' = 0$

$$\Rightarrow (\mu - 1)A - (\mu' - 1)A' = 0$$

or $\boxed{(\mu - 1)A = (\mu' - 1)A'}$



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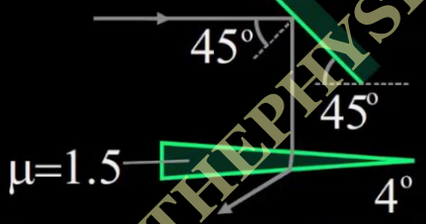
→ This formula can also be extended to achieve dispersion without deviation with n prisms. Then our formula is modified to:

$$\boxed{\Sigma (\mu_i - 1) A_i = 0}$$

→ Here A_i is a signed value and considered negative for inverted prisms.

EXAMPLE:

A ray of light strikes a plane mirror at an angle of incidence 45° as shown in the figure. After reflection, the ray passes through a prism of refractive index 1.5, whose apex angle is 4°. The angle through which the mirror should be rotated if the total deviation of the ray is to be 90° is:

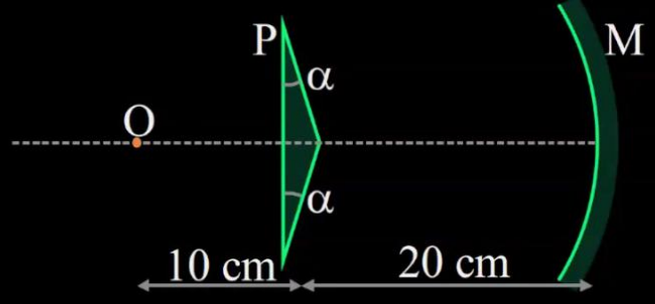


(A) 2.0° ↺ (B) 2.0° ↻ (C) 1.0° ↺ (D) 1.0° ↻

Solution: From mirror itself the ray was bent 90°. Now prism causes an additional deviation of $\delta = (\mu - 1) A = 0.5 \times 4° = 2°$ clockwise. Thus mirror should be rotated 1° anticlockwise. (C) Ans.

EXAMPLE:

O is the point object kept on the principal axis on a concave mirror M of radius of curvature 20 cm. P is a prism of angle $\alpha = 1.8°$? Light falling on the prism and then fall on the mirror. Find the distance between the images formed by the concave mirror due to this object. Take $\mu = 3/2$ for prism.



(A) π/10 cm (B) π/20 cm (C) π/30 cm (D) π/40 cm

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Solution: The biprism forms two images of O at I distance

$h = 10\text{ cm} \times \left(\frac{3}{2} - 1\right) \times 1.8 \times \frac{\pi}{180} = \frac{\pi}{20}$ cm (Above & below the object).

Now magnification, $m = f/x = 10/20 = \frac{1}{2}$ ⇒ image distance of final images from principal axes,

$h' = \frac{1}{2} \times \frac{\pi}{20} = \frac{\pi}{40}$ cm (Inverted images). Thus separation between the two images formed by concave mirror: $\frac{\pi}{40} \times 2 = \frac{\pi}{20}$ cm i.e option (B).

Note: These are not the final images. For example consider the adjacent ray diagram. There will be further procesing of reflected rays from the prisms, and another shifting will happen towards apex of the prism.

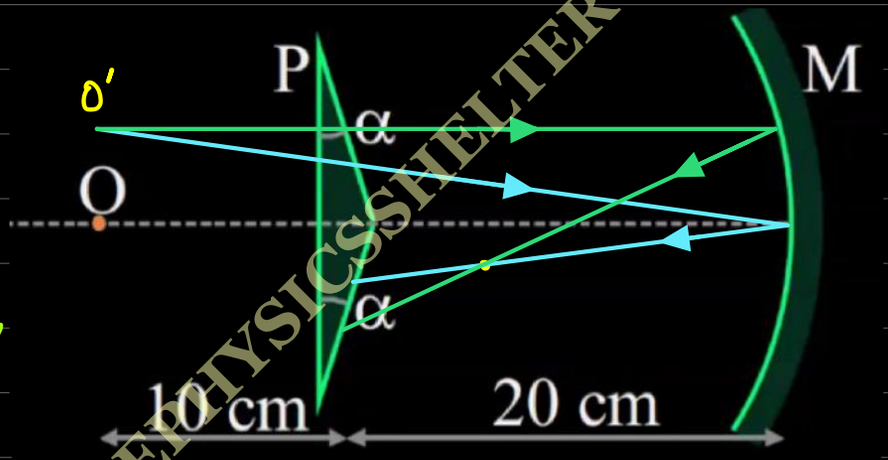
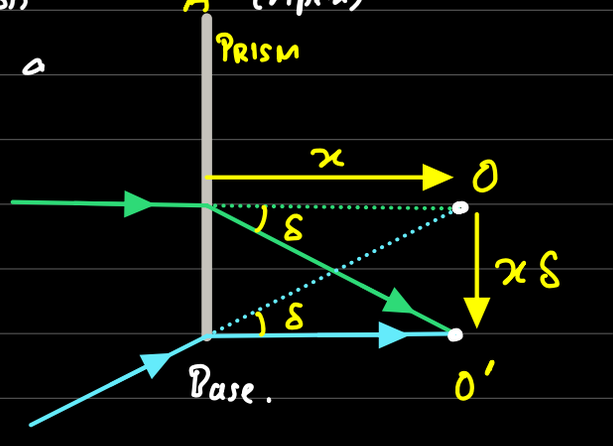


Image Formation From A Thin Prism For Virtual Object:

The adjacent figure shows ray diagram for image formation from a thin prism for a virtual object. Notice that with real object, image was shifted towards the apex, while with virtual object image is shifted towards the base.



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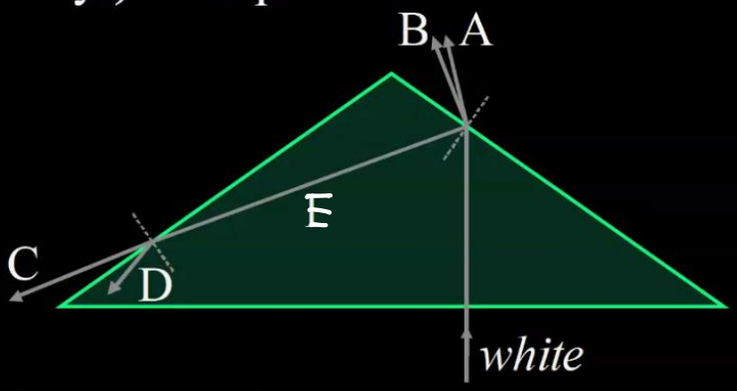
EXAMPLE:

A white light ray is incident on a glass prism and it creates three refracted rays A, B & C and one reflected ray D. Select the correct alternative(s). (E & D are internally reflected rays). Best possible colour for



(A) (A) is red, (B) is green, (C) is yellow and (D) is blue
(B) (A) is red, (B) is yellow, (C) is green and (D) is blue
(C) (A) is blue, (B) is green, (C) is yellow and (D) is red
(D) (A) is blue, (B) is yellow, (C) is green and (D) is red

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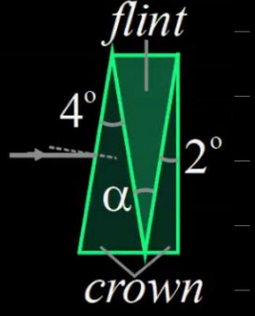
Solution: D is undergoing TIR at both interfaces so it must have highest refractive index. C undergoes TIR at first interface but not A and B, so it has more refractive index than A and B. B bends more strongly than A, so it has more refractive index than A. Thus correct order of refractive indices:

$\mu_D > \mu_C > \mu_B > \mu_A$, among VIBGYOR, V has highest μ and R has lowest. Thus only correct option possible is B.

EXAMPLE:

The refractive indices of the crown glass for violet and red lights are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. A narrow beam of white light is incident at a small angle of incident on the shown combination of thin prisms. The value of α is chosen such that the mean deviation of the incident beam is zero.



(A) $\alpha = 4^\circ$
(B) dispersion $= 0.04^\circ$
(C) $\alpha = 2^\circ$
(D) dispersion $= 0.02^\circ$

Solution: Mean deviation: $\delta_y = \Sigma (\mu_i - 1) A_i$. We may assume that refractive index for yellow color is midway between that of violet and red. Thus $\mu_c = 1.5$ and $\mu_f = 1.75$ [c: crown f: flint]
Also recall that A_i are signed values. Thus:

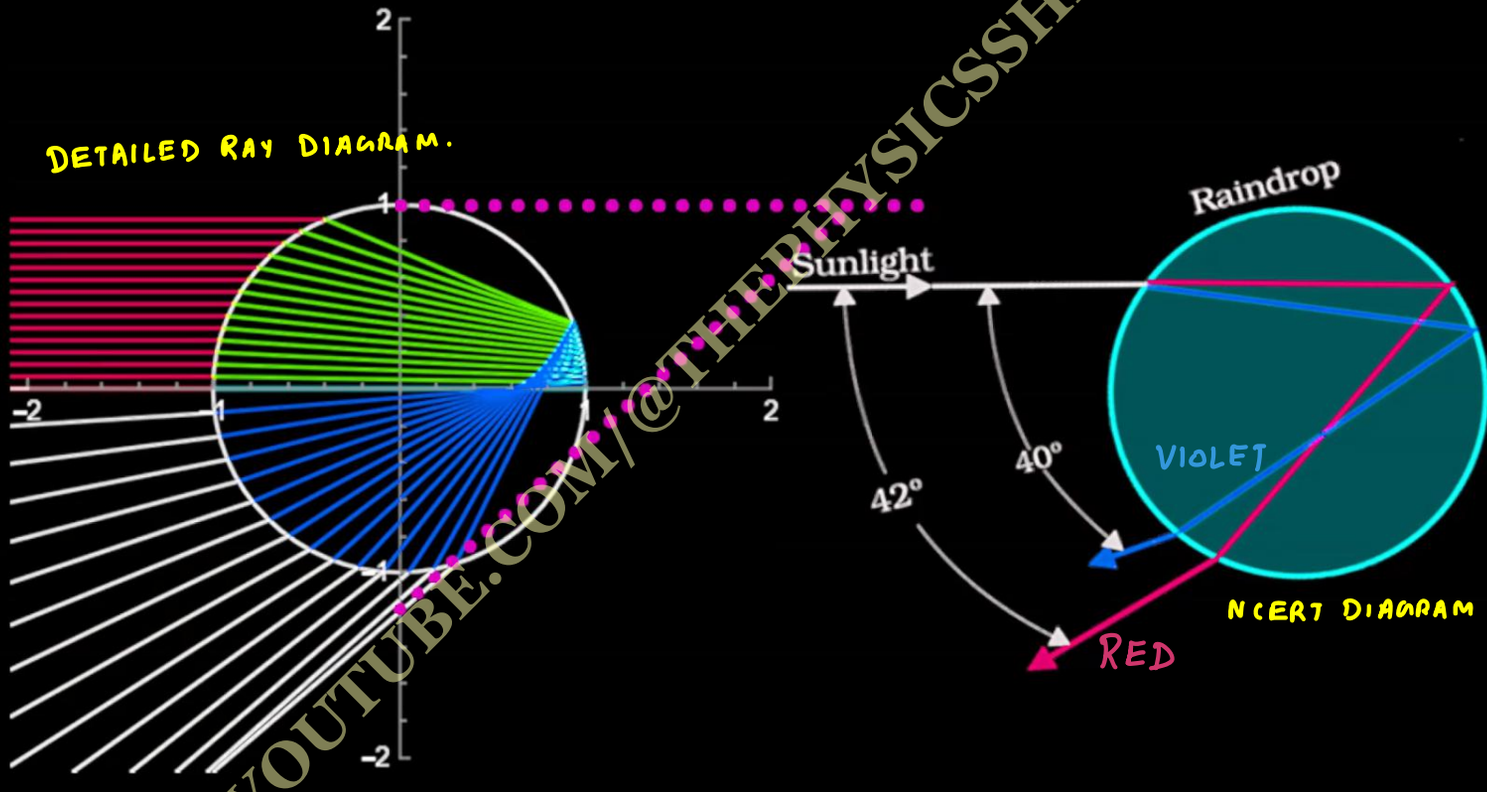
$$0.5 \times 4 - 0.75\alpha + 0.5 \times 2 = 0 \Rightarrow \alpha = 4^\circ \text{ Ans.}$$

Now dispersion:

$$\phi = \Sigma (\mu_{v_i} - \mu_{r_i}) A_i = 0.02 \times 4 - 0.04 \times 4 + 0.02 \times 2$$
$$= -0.04^\circ \text{ Ans.}$$

Thus correct option: A and B Ans.

PRIMARY RAINBOW FORMATION:



A primary rainbow is formed due to two refractions and one partial internal reflection as shown in the ray diagram above. For ease of understanding consider only red component of sunlight. As can be seen from the figure, after being processed from the raindrop, the red rays are going in several directions, however their density is not same at every angle. It can be mathematically shown that for horizontal incidence of sunlight, there is maximum crowding of red rays at an angle of 42° (see NCERT figure). and similarly violet light is most crowded at a 40° angle. AMBARISH SIR

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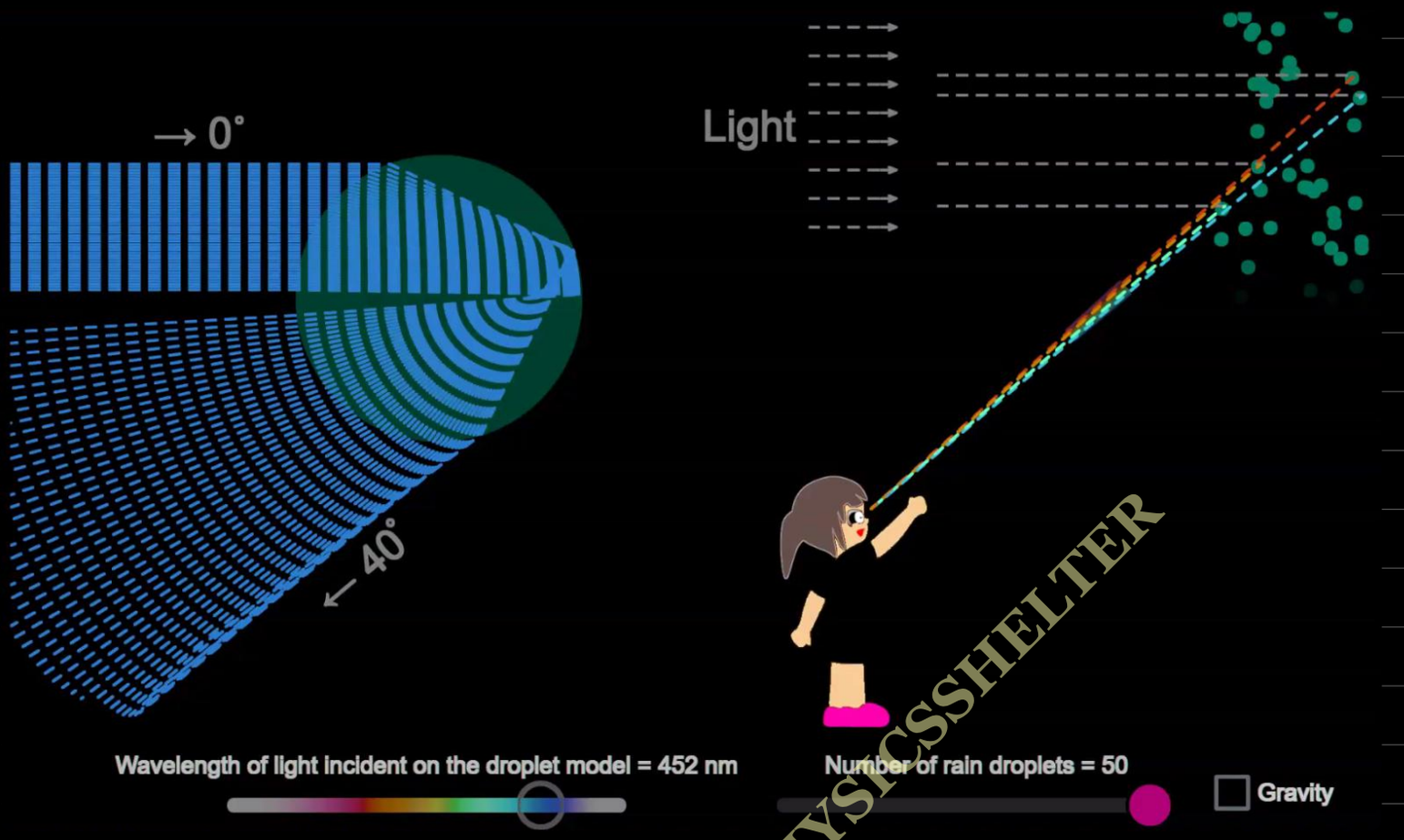

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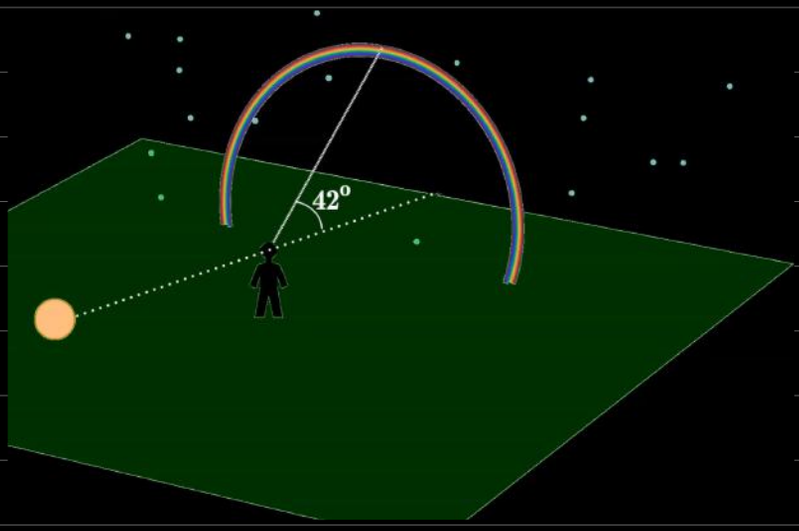
Now there are several such drops and each drop sending a red light 42° below the horizon thus when a person looks 42° above the horizon, red light from the drops along line of sight falls in to the person's eyes, similarly looking at an elevation of 40°, one sees violet light.

CAUTION: Don't get confused, because the red beam is the lower beam after being processed by the drops, because of this very reason, red appears to be the highest color in the primary rainbow. Remember that you are not simultaneously seeing both beams from a single drop. At a given elevation you are seeing only certain colors being emanated at that elevation from the drops that are present at the line of sight. When you are looking at a different elevation, you are looking at different drops. (See the above figure carefully).

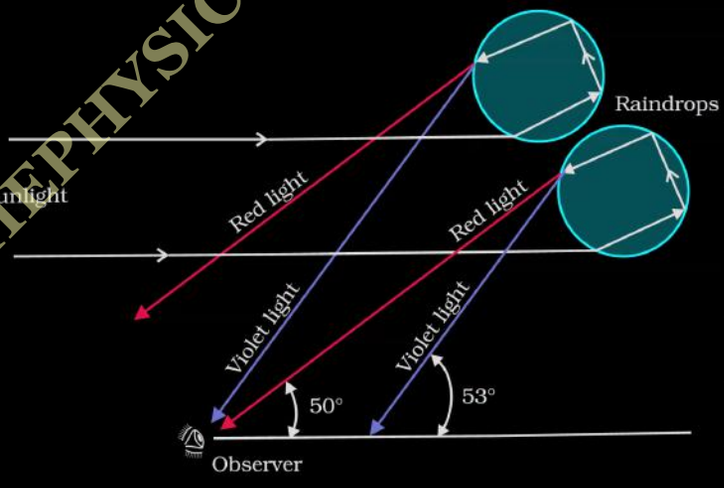
→ In a primary rainbow, red is at the top (42° elevation) Violet is at the bottom (40° elevation). From a particular drop, red rays are lower than the violet rays.

Realize that there is a perfect cylindrical optical symmetry about the line joining the sun to the head of the observer, thus similar optical phenomena is occurring along a circular arc as shown in the figure and this is why rainbow appears to be of circular arc shape. (Rainbow would appear completely circular if our view were not blocked by the horizon and water droplets were present there.



SECONDARY RAINBOW FORMATION:

Just like primary rainbows, secondary rainbows are also formed due to refraction and partial reflections from raindrops, however there, we have two partial reflections. In secondary rainbows, violet color appears at a greater elevation (53°) and red color appears lower (50°). Further, intensity is lost at every reflection/transmission thus secondary rainbows are very faint as compared to primary rainbows and most of the times almost invisible under normal atmospheric conditions. However one can artificially see such rainbows in labs / waterparks under suitable lighting conditions.



A falling water drop in suitable lighting condition will also appear to change colors from violet to red and then back to violet. This also happens due to same physical phenomena that is responsible for secondary and primary rainbow formation and it is illustrated on the next page.

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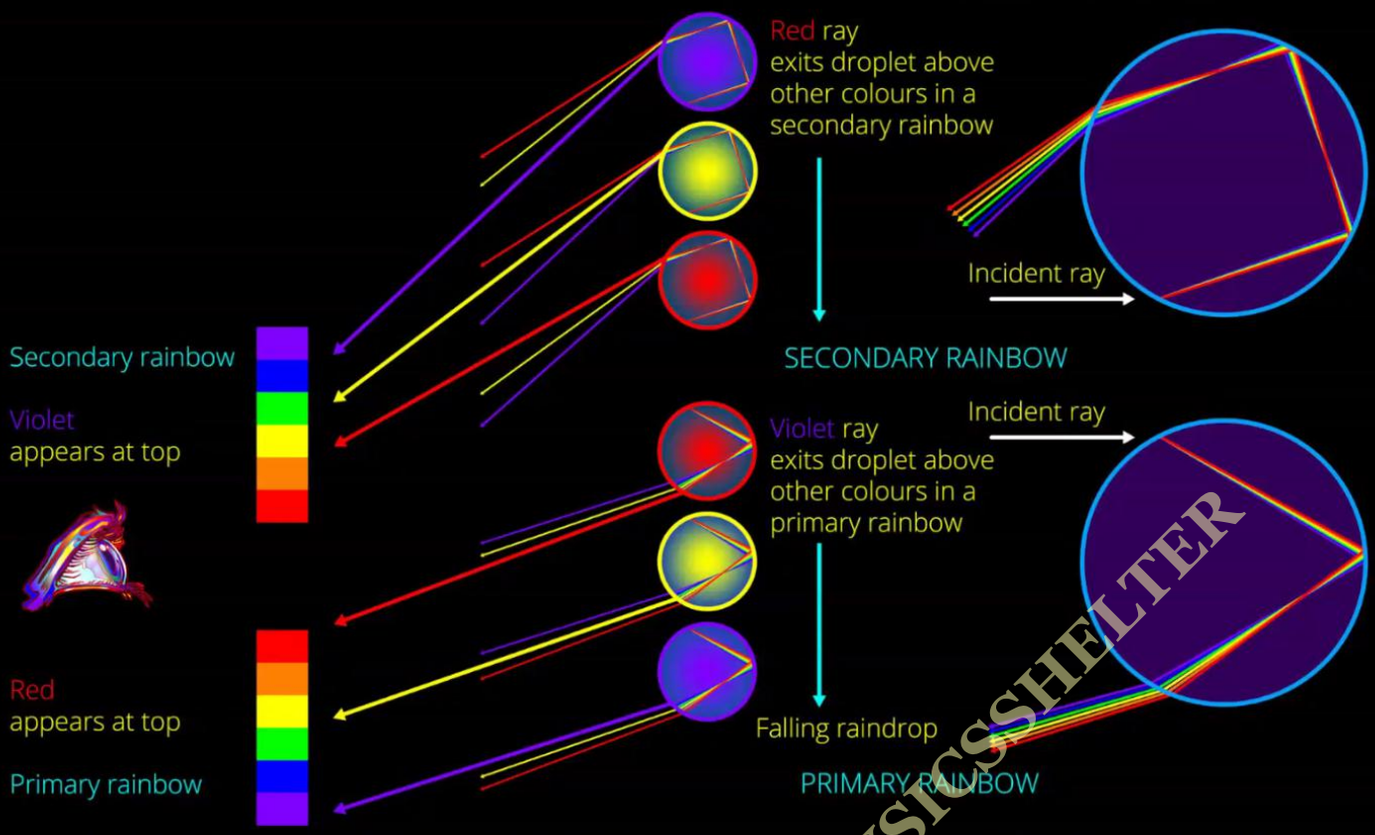
CHANGING COLOUR OF A FALLING RAINDROP



A single falling raindrop can contribute to the formation of a secondary and then a primary rainbow.
The position of the droplet relative to the observer determines its colour.

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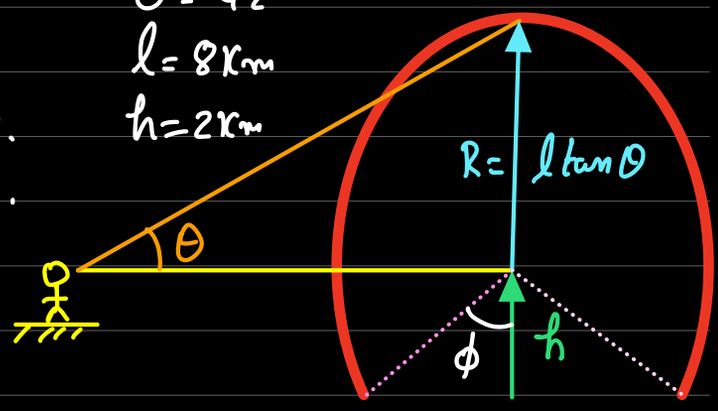
EXAMPLE: VISIBLE FRACTION OF RAINBOW ARC:

A mountaineer on a peak observes a rainbow in a valley, caused by water droplets in the air 8 km away. The valley is 2 km below the mountain peak and entirely flat. What fraction of the complete circular arc of the rainbow is visible to him ? Given line joining the sun to the head of the person is perfectly horizontal.

Solution: Consider the schematic diagram shown in the figure. Here the radius of the rainbow arc is $l\tan\theta$. We are interested in finding $360° - 2\phi$.

$\theta = 42°$
$l = 8\,km$
$h = 2\,km$



Now $\cos\phi = \frac{h}{R} = \frac{h}{l\tan\theta}$

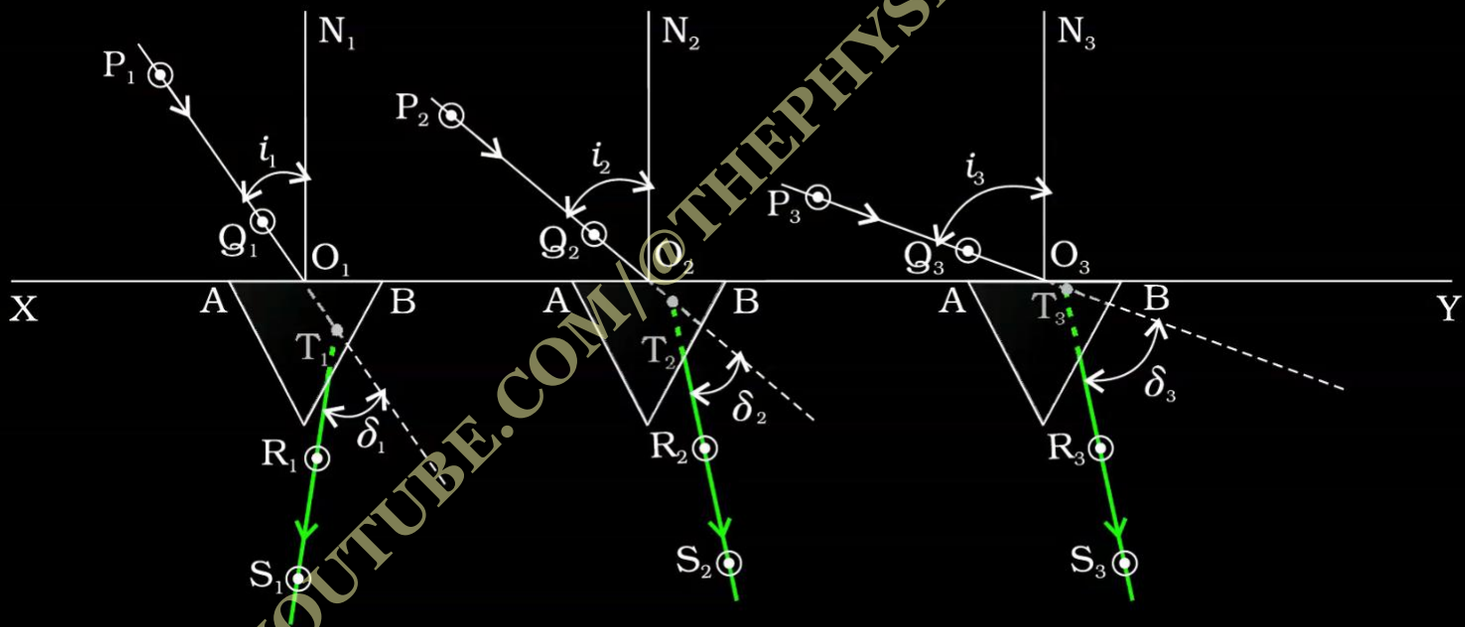
$\Rightarrow \cos\phi = \frac{2\,km}{8\,km \times \tan 42°} = 0.278$

$\Rightarrow \phi = 73.9°$. Thus visible angle of rainbow arc $= 360° - 2\times 73.9°$

$\Rightarrow$ Visible arc angle $= 212.24°$ Ans. Visible fraction $= 212.24°/360°$

$\Rightarrow f \approx 0.6$ Ans.

Experiment To Determine Minimum Angle Of Deviation From A Prism:



For doing this experiment we take a drawing sheet and draw a line XY on it and mark several points O_1, O_2, O_3 etc at a distance of 8 to 10 cm from each other for placing the prism in various trials. Now we draw normal to XY from each point O_1, O_2 etc. Now we choose an angle of incidence i_1 for the first trial, we draw the angle i_1 from the Normal and affix two allpins P_1 and Q_1 to indicate the path of the incident ray. Now we place the prism with one of its equilateral arms coinciding with XY and

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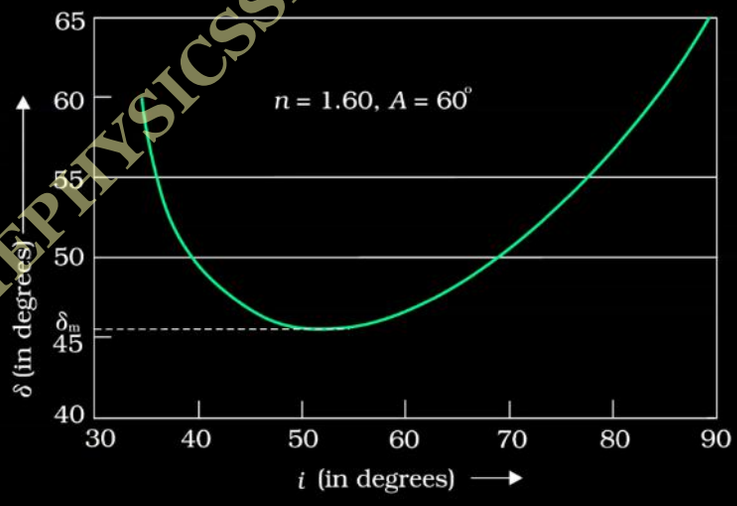
Refraction happening at O_1. We look at the allpins from the other side of the angle B (see figure) from a view point where the two allpins appear collinear (eclipsed). Now we fix two more allpins on the drawing sheet (affixed to drawing board) to represent the refracted ray (these allpins eclipse the images of allpins on the incident side), let us call them R_1 and S_1. Now we draw a line connecting R_1 and S_1 to represent the refracted ray. We then measure the angle between incident ray and refracted ray to find the angle of deviation. The experiment is again repeated with different angles of incidence (between 40° and 70° at intervals of 5°). Now we have a table of incidence angle vs deviation and we can use it to plot the graph of deviation vs incidence angle.

After plotting all the data points, we fit a smooth curve through them and then draw a tangent at lowest point to find the minimum deviation. Refractive index of the prism is now calculated as:

$$\mu = \frac{\sin[(A+\delta_m)/2]}{\sin(A/2)}$$



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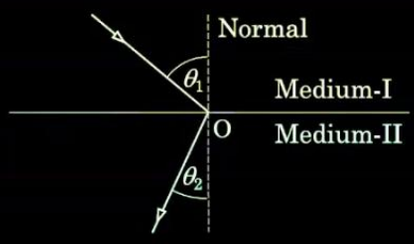
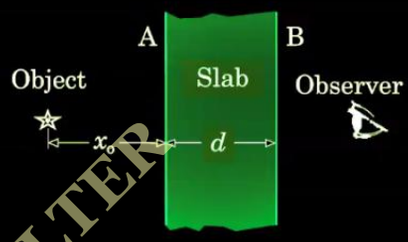
EXAMPLE: PATHFINDER: MCQ1S:

If Snell's law is found valid for two mediums, whose refractive indices have opposite signs, the incident and the refracted rays must remain on the same side of the normal at the point of incidence. In the figure is shown a ray crossing the interface between two mediums I and II. The ray makes angles θ_1 and θ_2 with normal in the medium-I and II. The refractive indices of the mediums μ_1 and μ_2 have opposite signs.



Snell's law is expressed for the above situation as usual by the equation $\mu_1 \sin\theta_1 = \mu_2 \sin\theta_2$. Therefore, to satisfy this equation, the angles θ_1 and θ_2 must also have opposite signs.

A self-luminous point object placed at a distance x_0 from a slab of transparent material of negative refractive index is viewed through the slab as shown in the figure. Thickness of the slab is d, refractive index of material of the slab with respect to the outside medium is $\mu = -1$.



For different moduli of values of x_0 and d the observer may find real or virtual image of the object. Which of the following statements is true?

(a) If $x_0 > d$, image is virtual and for $x_0 < d$ image is real.
(b) If $x_0 < d$, image is virtual and for $x_0 > d$ image is real.
(c) If $x_0 \le d$, image is virtual and for $x_0 > d$ image is real.
(d) If $x_0 > d$, image is virtual and for $x_0 \le d$ image is real.

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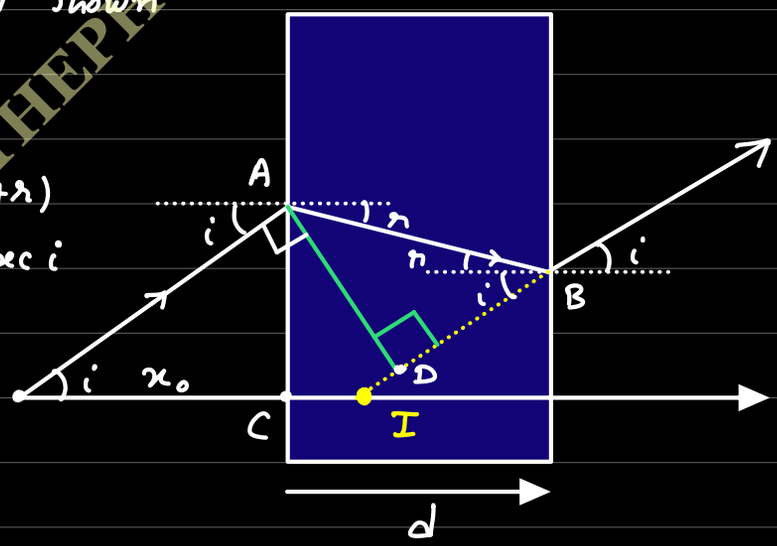
Solution: Consider the ray diagram shown in adjacent figure.

$AB = d\sec r$

$AD = AB \sin(i+r) = d\sec r \sin(i+r)$

$OI = AD \operatorname{cosec}(i) = d\sec r \sin(i+r) \operatorname{cosec} i$

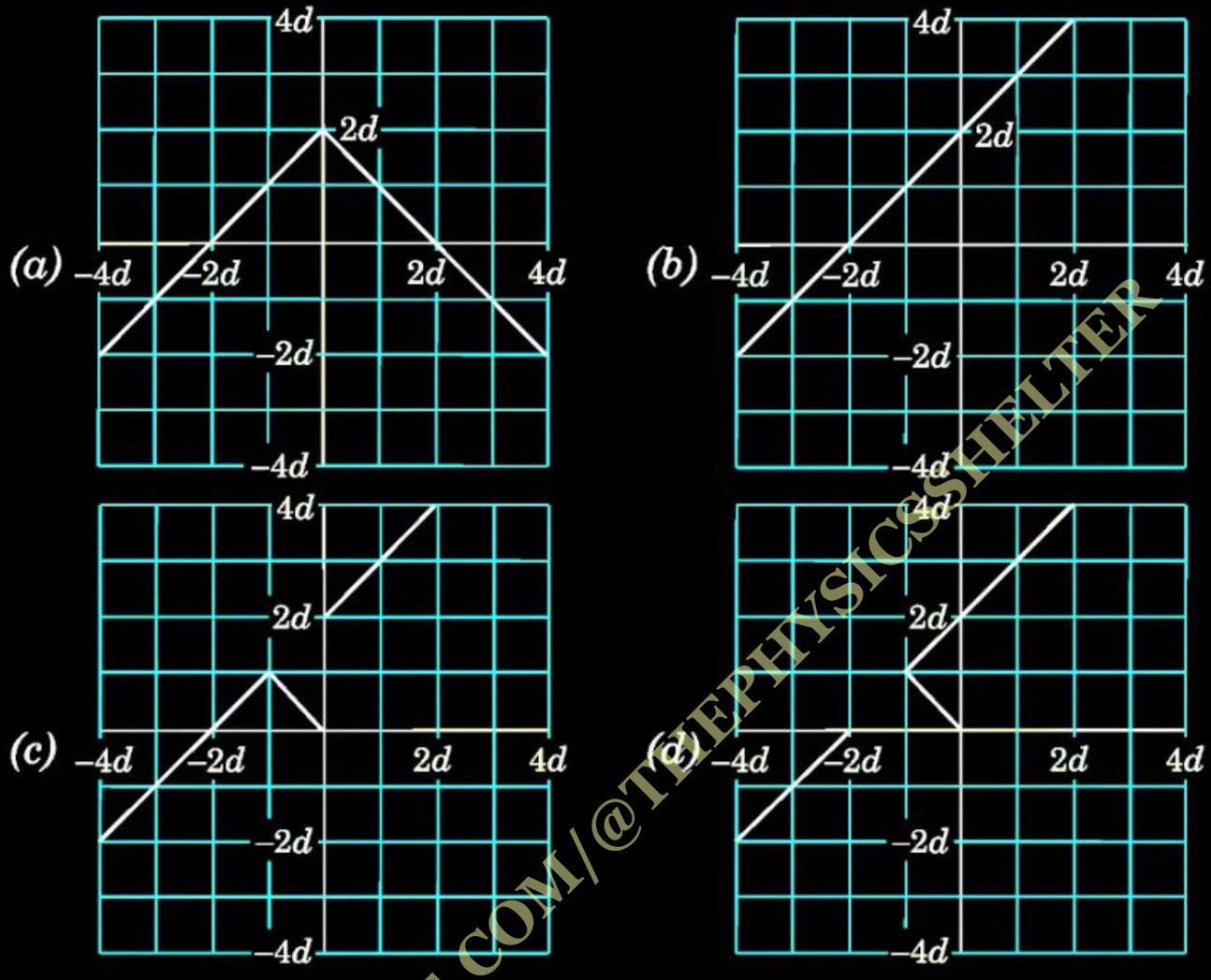
Now $\because \mu = -1$, $i = r$ (direction is already taken in to account so no need to write $-r$). Thus,



$$OI = \frac{d \sin 2i}{\sin i \cos i} = 2d$$ i.e image is formed of a distance $2d$ from the object thus if $2d > x_0 + d$ then image is formed after crossing the slab and will be therefore real, thus for real image, $x_0 < d$ and for virtual image $x_0 > d$. Option (a) ans.

Example : Pathfinder : MCQ17:

Distances x_o and x_i of the object and the final image are measured from the face A of the slab. The Cartesian sign conventions are also followed. Which one of the following graphs correctly represents relationship between x_o as abscissa and x_i as ordinate?



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Solution: Ray diagram for real object was shown in the last example and we found that image was shifted through a distance 2d in direction of incident light. Thus, $x_i = x_o + 2d$. For virtual objects we can use the concept of reversibility. Thus in the same ray diagram we can reverse the rays and take its mirror image. Thus for virtual objects also, the image is shifted by a distance 2d in direction of incident light. Thus for virtual objects also, $x_i = x_o + 2d$. This gives us the correct graph i.e (b) Ans.